

MICRO CAT Rev 1 — printed part reference drawings

Actuated chibi cat; 14 policy joints + jaw. Millimetres. Design pose (straight legs). Assembly frame: X forward, Y left, Z up, floor Z=0, trunk frame Z=120.

20 printed parts (474.7 g solid PLA equivalent) + 18 purchased items (399.0 g catalogue).

Overall views generated from the actual meshes; use STEP for exact geometry. Not production GD&T; prototype fit and fastener checks are still required.

Printed parts

left_yaw2roll	link left_yaw2roll	32.4 x 26.8 x 40.4 mm	17.0 g
left_hip	link left_hip	33.5 x 38.1 x 29.3 mm	6.3 g
left_upper_leg	link left_upper_leg	62.2 x 31.8 x 44.5 mm	23.1 g
left_lower_leg	link left_lower_leg	26.8 x 14.4 x 62.5 mm	8.6 g
left_foot	link left_foot	90.0 x 46.0 x 38.0 mm	38.7 g
right_yaw2roll	link right_yaw2roll	32.4 x 26.8 x 40.4 mm	17.0 g
right_hip	link right_hip	33.5 x 38.1 x 29.3 mm	6.3 g
right_upper_leg	link right_upper_leg	62.2 x 31.8 x 44.5 mm	23.1 g
right_lower_leg	link right_lower_leg	26.8 x 14.4 x 62.5 mm	8.6 g
right_foot	link right_foot	90.0 x 46.0 x 38.0 mm	38.7 g
chassis	link trunk	131.0 x 62.0 x 46.5 mm	44.2 g
torso_shell	link trunk	86.0 x 77.2 x 47.8 mm	29.1 g
tail_cover	link trunk	81.1 x 66.4 x 81.6 mm	42.6 g
neck_link	link neck	24.8 x 31.4 x 71.7 mm	16.0 g
neck_sleeve	link neck	31.6 x 36.8 x 60.4 mm	17.2 g
head_base	link head_base	26.0 x 35.0 x 25.1 mm	4.4 g
head_yoke	link head_yoke	31.3 x 37.8 x 32.4 mm	15.4 g
head_frame	link head	84.2 x 61.9 x 41.7 mm	25.1 g
skull	link head	108.0 x 112.0 x 85.9 mm	86.5 g
jaw_chin	link jaw	58.0 x 40.0 x 32.0 mm	7.0 g

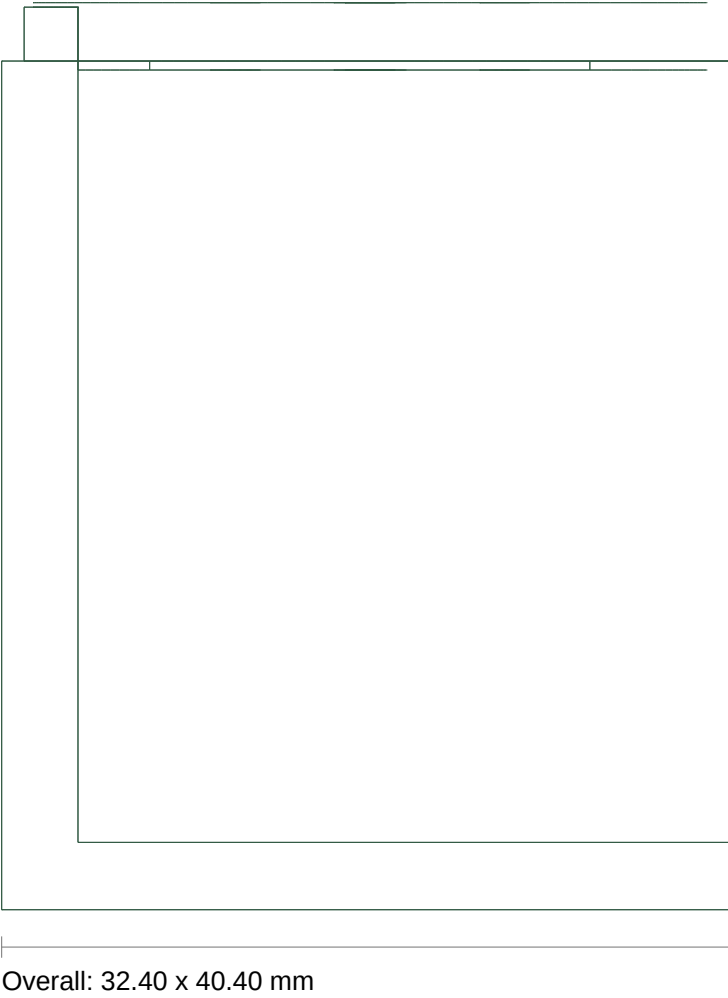
Purchased items (envelopes)

servo_left_hip_yaw	actuator	link trunk	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_left_hip_roll	actuator	link left_yaw2roll	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_left_hip_pitch	actuator	link left_upper_leg	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_left_knee	actuator	link left_upper_leg	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_left_ankle	actuator	link left_lower_leg	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_right_hip_yaw	actuator	link trunk	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_right_hip_roll	actuator	link right_yaw2roll	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_right_hip_pitch	actuator	link right_upper_leg	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_right_knee	actuator	link right_upper_leg	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_right_ankle	actuator	link right_lower_leg	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_neck_pitch	actuator	link neck	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_head_pitch	actuator	link neck	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_head_yaw	actuator	link head_yoke	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_head_roll	actuator	link head	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
servo_jaw	actuator	link head	18.0 g	XL330-class smart servo envelope; confirm the case hole pattern before
battery_pack	battery	link trunk	95.0 g	Removable 2S 18650 pack, 65 x 37 x 19 mm class, 7.4 V ~3000 mAh; slide
compute_board	electronics	link trunk	30.0 g	Single-board computer, compact bay: 26.5 x 47 mm. Boards: board to be
camera_module_3	camera	link head	4.0 g	Raspberry Pi Camera Module 3 Standard class, 25 x 24 mm board; the nos

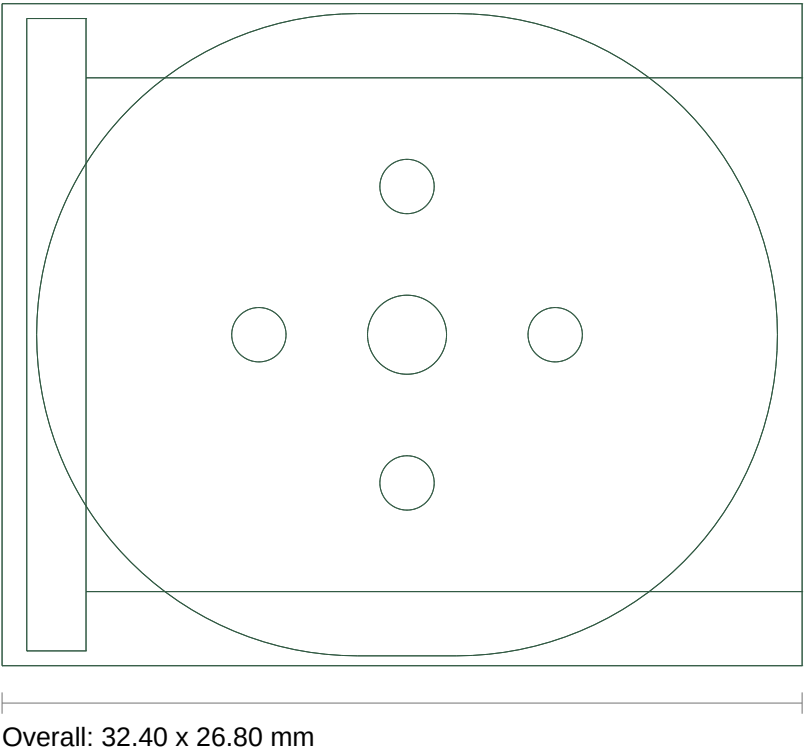
MICRO CAT Rev 1 / left_yaw2roll

link: left_yaw2roll | millimetres | original design, prototype reference, not production released

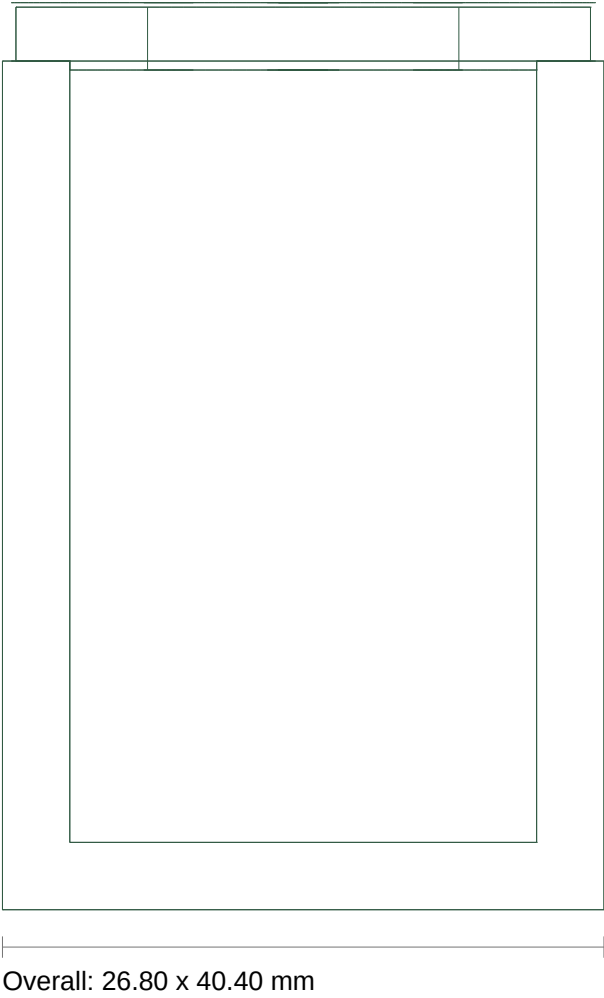
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 32.4 / 26.8 / 40.4 mm

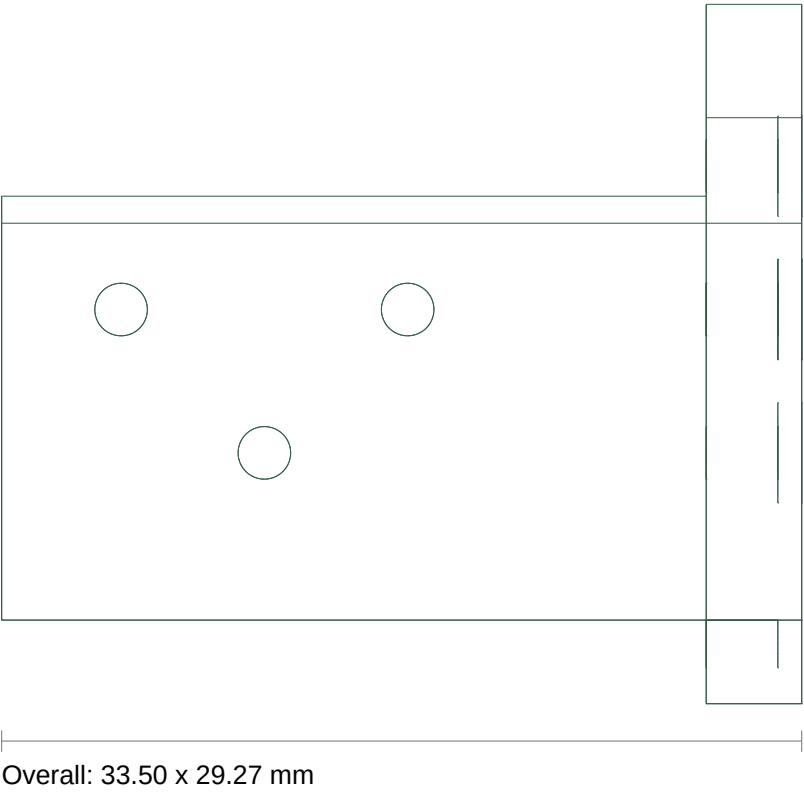
Material volume: 13731.2 mm3 | PLA solid-equivalent: 17.03 g (slicer mass will be lower)

Hip yaw horn plate on a U-channel around the hip roll servo. Print on the plate face; every wall is vertical.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

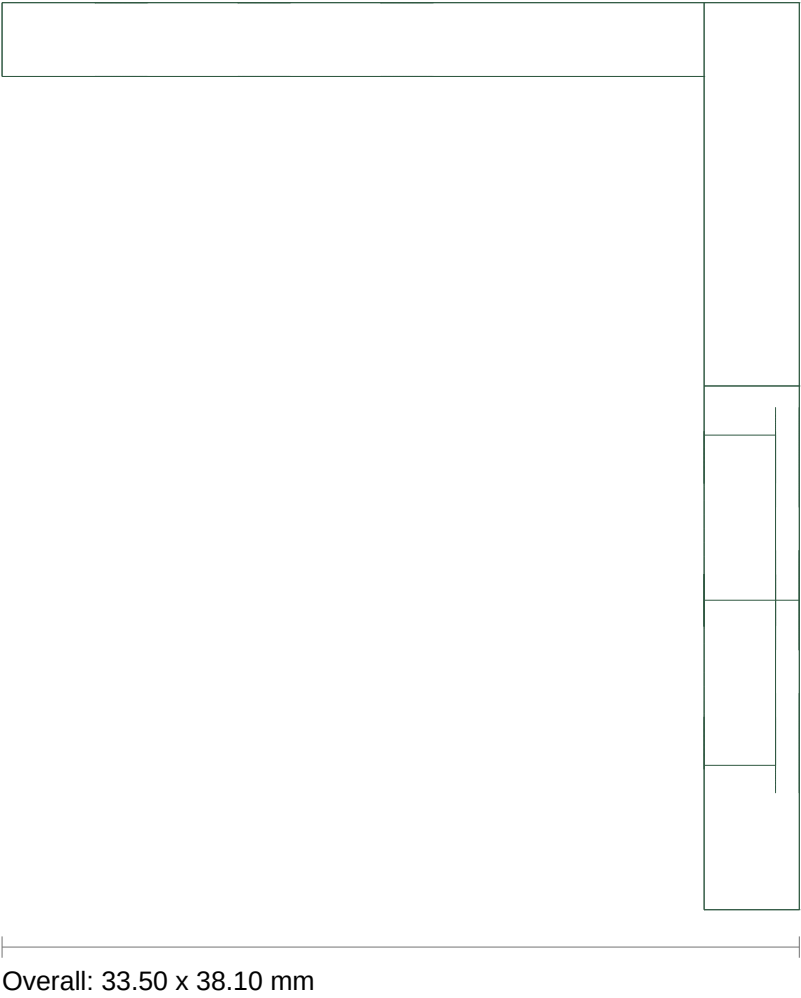
MICRO CAT Rev 1 / left_hip

link: left_hip | millimetres | original design, prototype reference, not production released

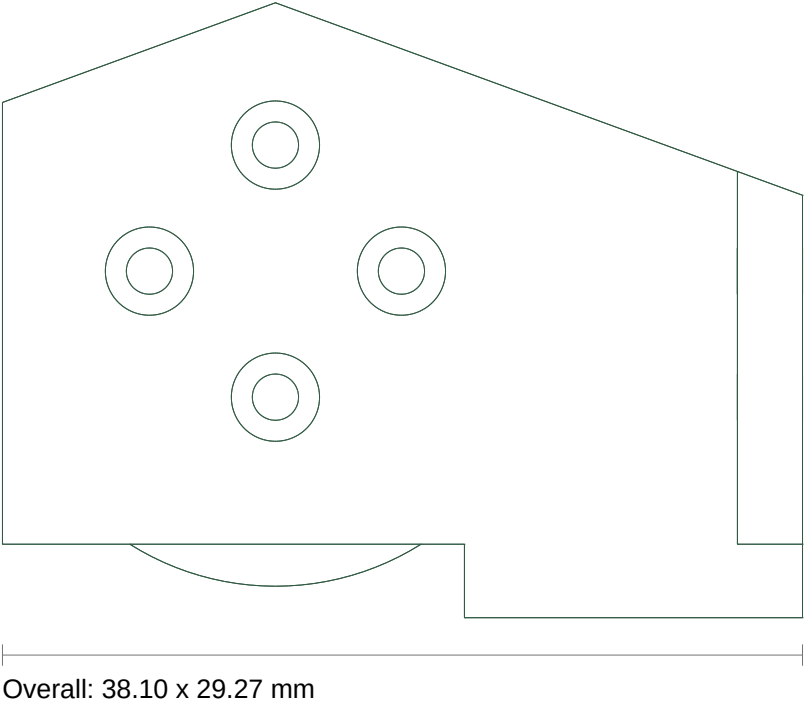
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 33.5 / 38.1 / 29.27 mm

Material volume: 5061.1 mm3 | PLA solid-equivalent: 6.28 g (slicer mass will be lower)

Hip corner bracket: roll horn plate and pitch horn plate joined by one rib. Print on the pitch plate face.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO CAT Rev 1 / left_upper_leg

link: left_upper_leg | millimetres | original design, prototype reference, not production released

SIDE X-Z



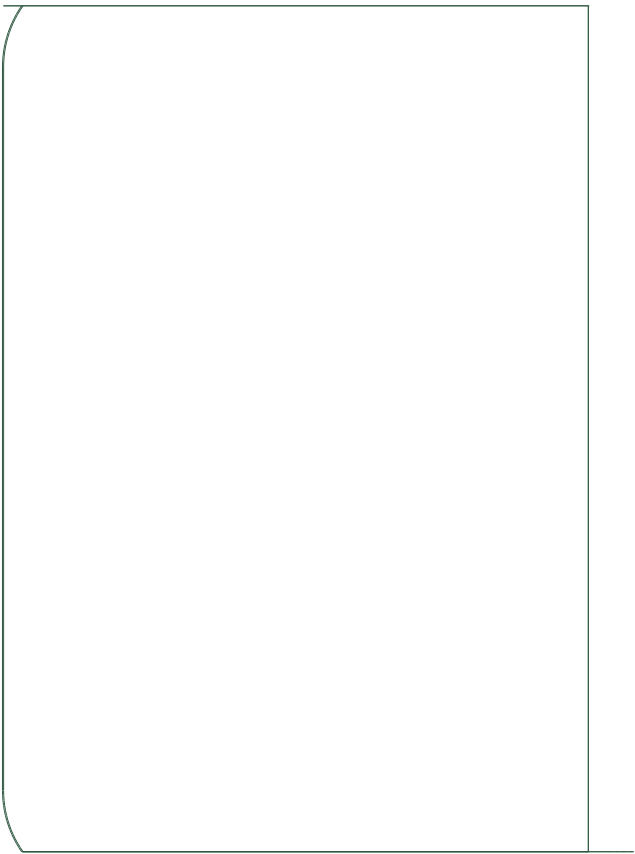
Overall: 62.20 x 44.50 mm

TOP X-Y



Overall: 62.20 x 31.75 mm

FRONT Y-Z



Overall: 31.75 x 44.50 mm

Assembly-frame extents X/Y/Z: 62.2 / 31.75 / 44.5 mm

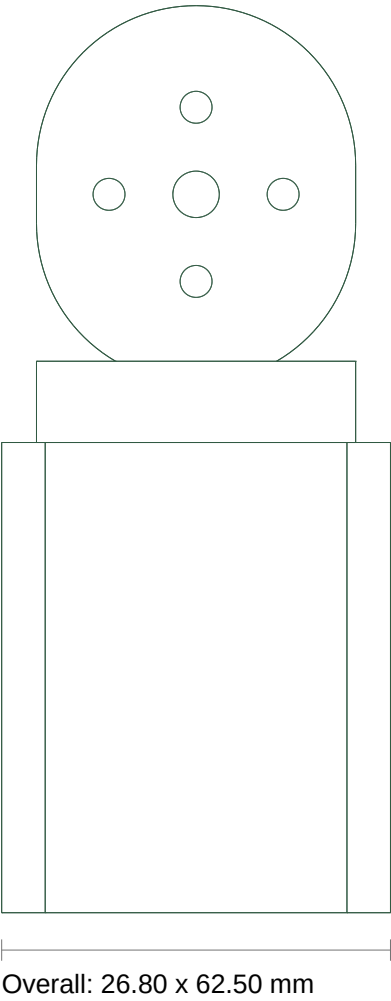
Material volume: 18593.5 mm3 | PLA solid-equivalent: 23.06 g (slicer mass will be lower)

Thigh shell around the hip pitch and knee servos, open on the inboard face. Print on the outboard face; the cavity opens upward, no support.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

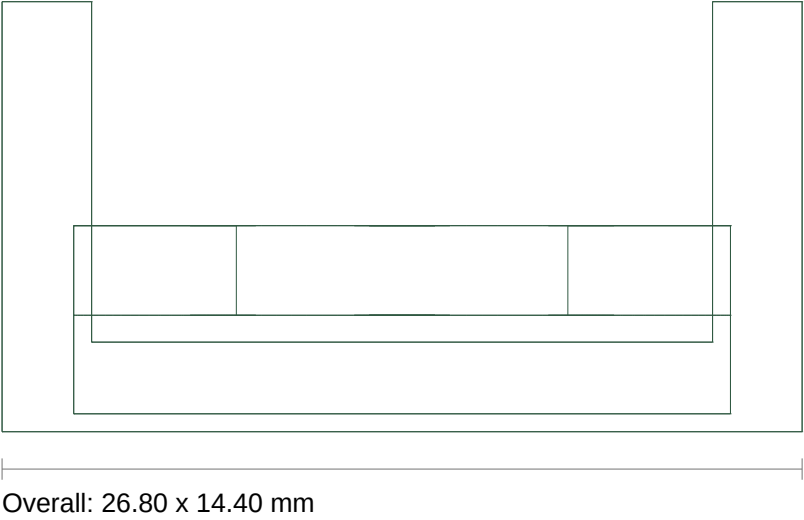
MICRO CAT Rev 1 / left_lower_leg

link: left_lower_leg | millimetres | original design, prototype reference, not production released

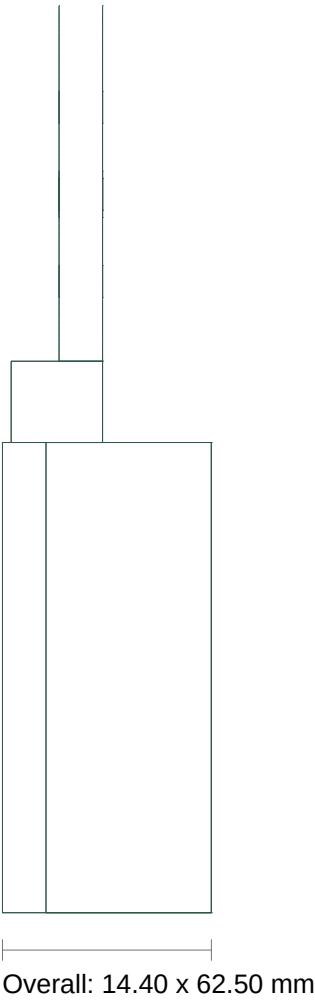
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 26.8 / 14.4 / 62.5 mm

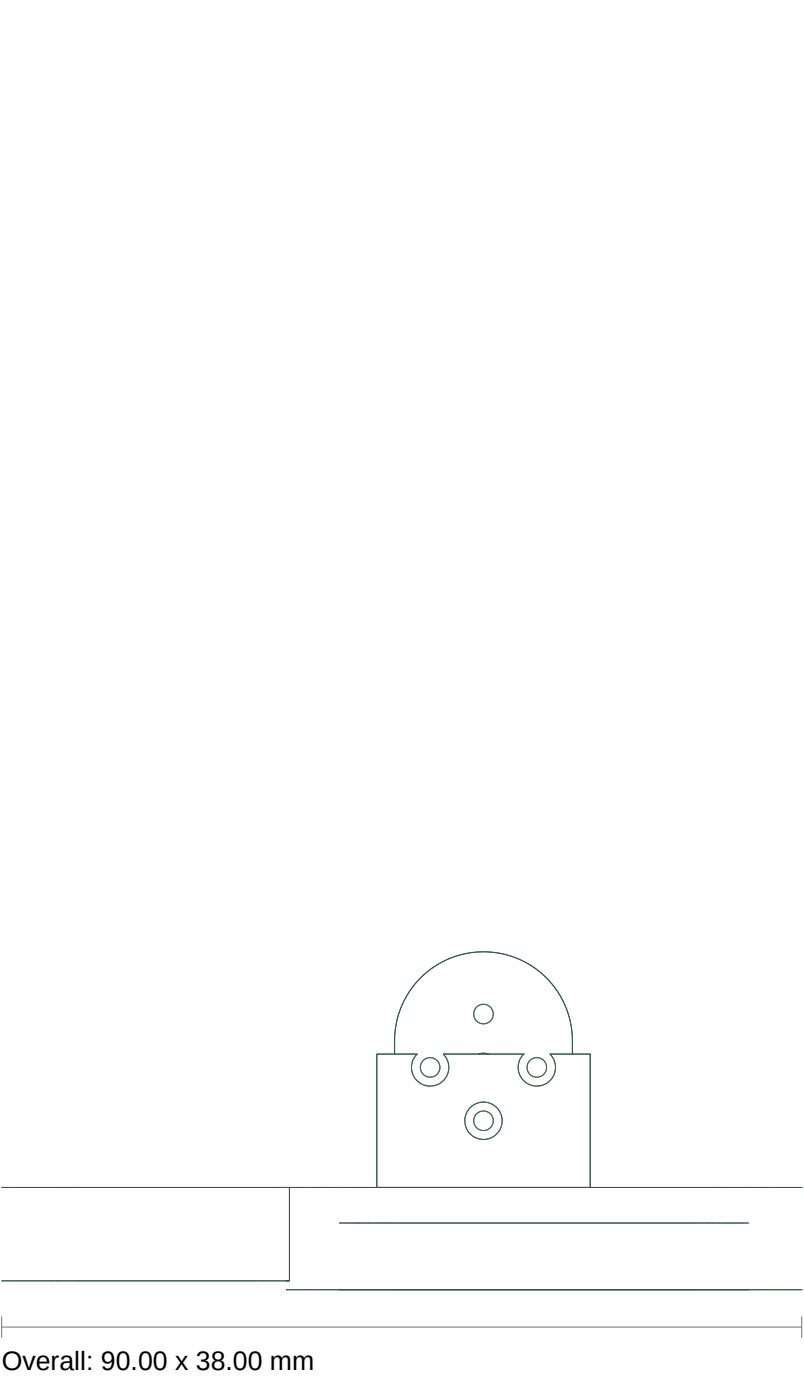
Material volume: 6898.7 mm3 | PLA solid-equivalent: 8.55 g (slicer mass will be lower)

Shin: knee horn plate stepping down into a channel around the ankle servo. Print on the inboard face.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

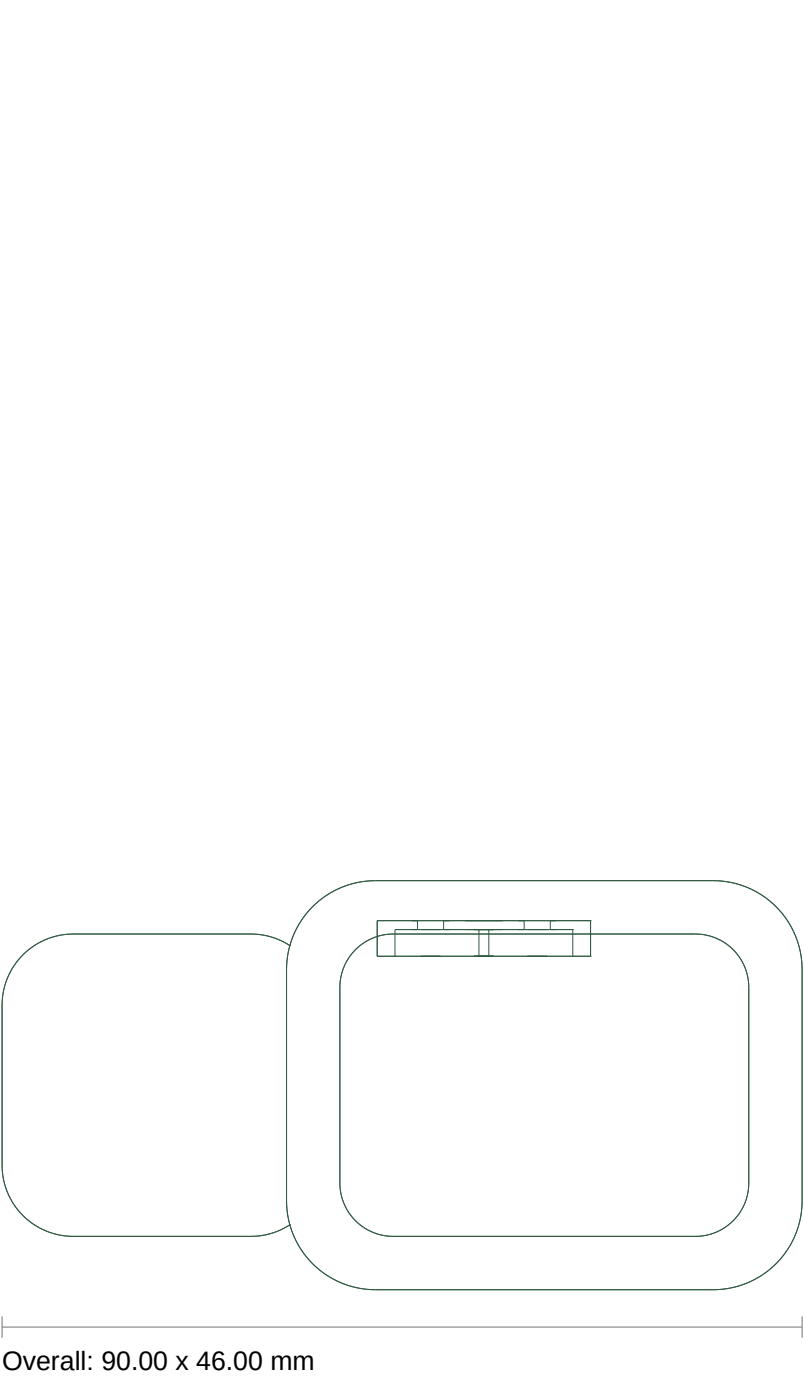
MICRO CAT Rev 1 / left_foot

link: left_foot | millimetres | original design, prototype reference, not production released

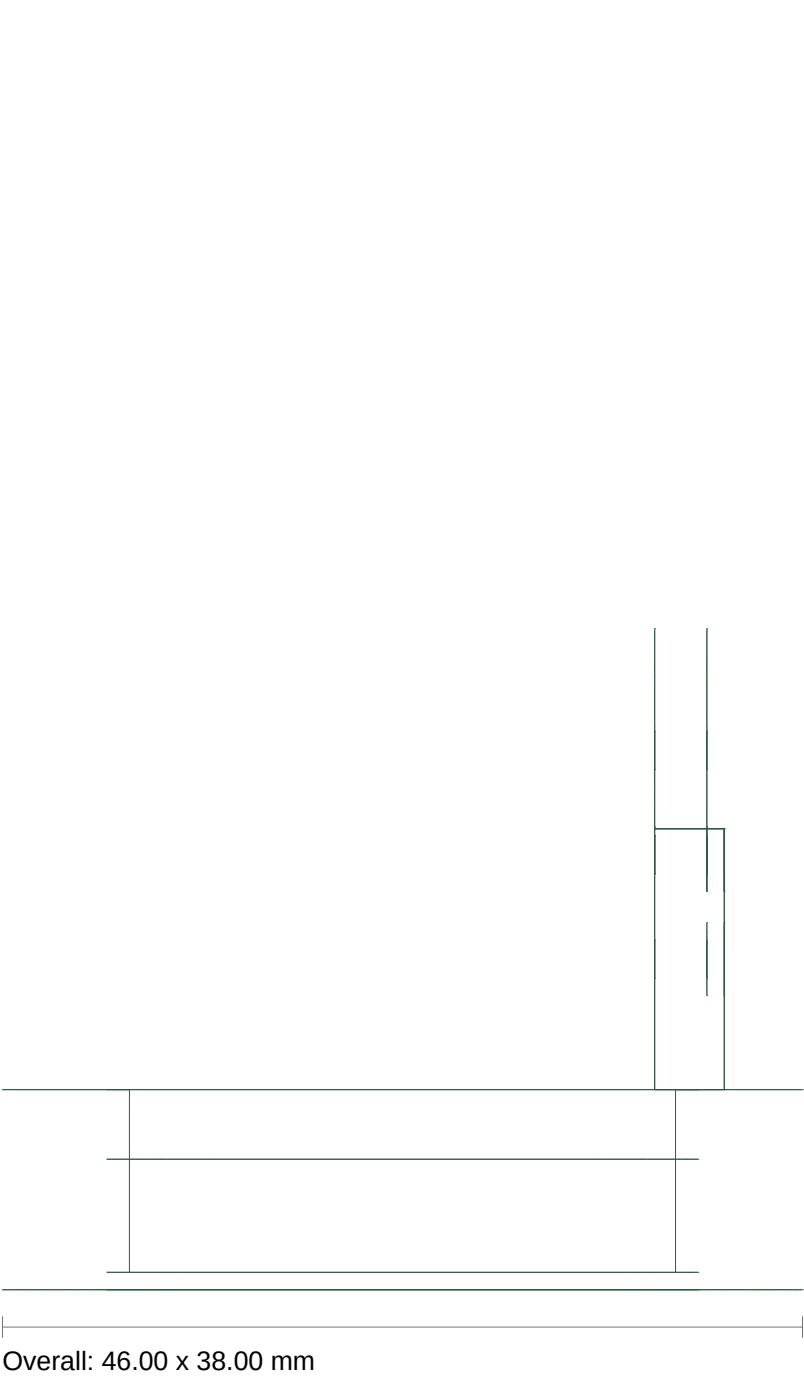
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 90.0 / 46.0 / 38.0 mm

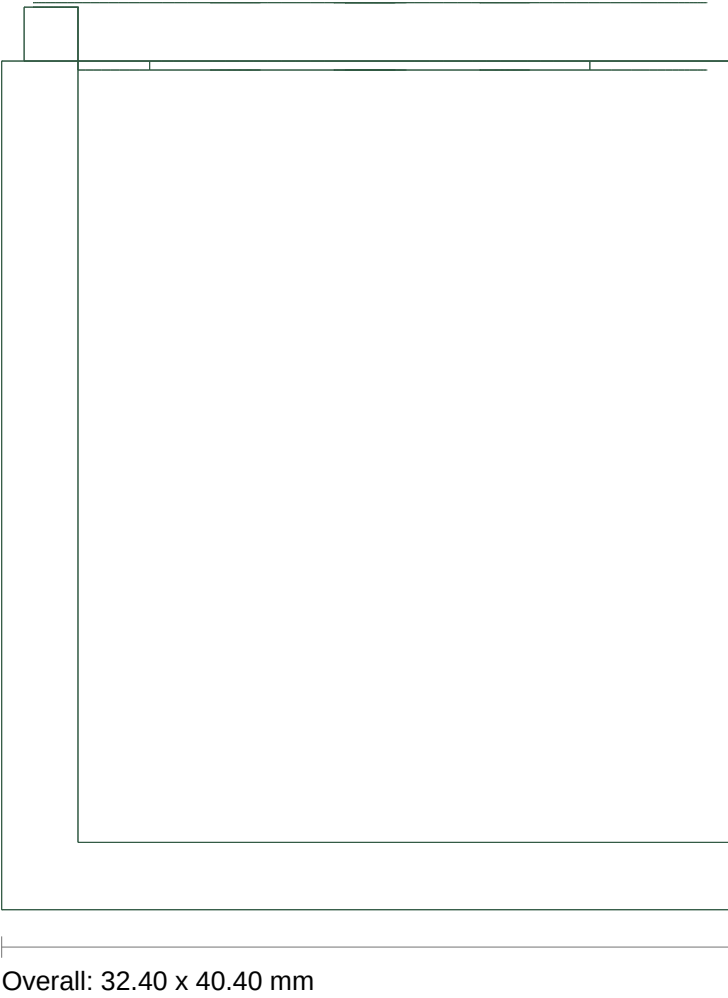
Material volume: 31227.4 mm3 | PLA solid-equivalent: 38.72 g (slicer mass will be lower)

Paw: ankle horn plate on a rounded sole, hollow underneath for a replaceable pad, plus a heel spur that sits 1.5 mm above the sole plane and reaches 24 mm behind it. Print sole down; the spur underside is a bridge 1.5 mm off the bed.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

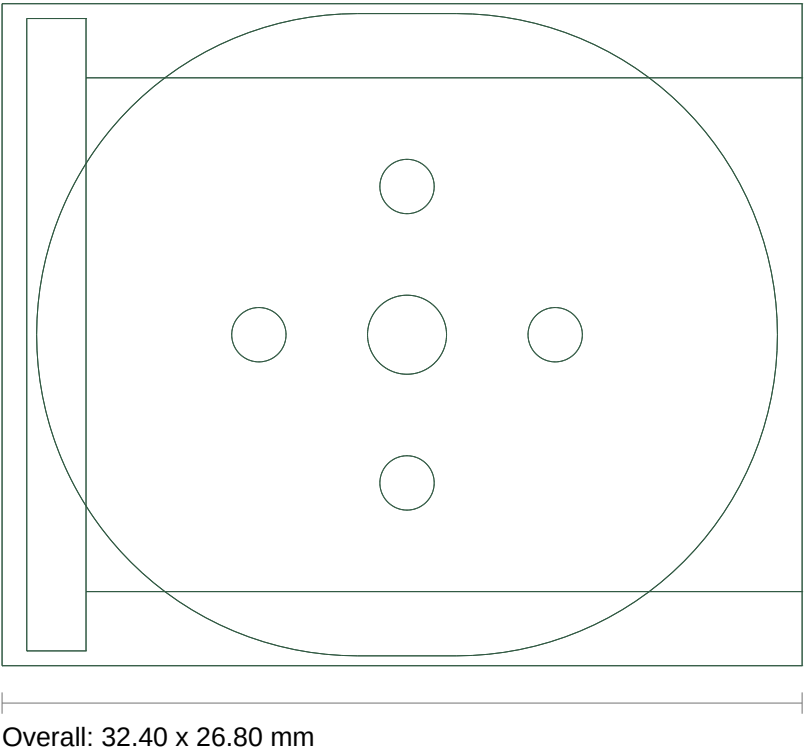
MICRO CAT Rev 1 / right_yaw2roll

link: right_yaw2roll | millimetres | original design, prototype reference, not production released

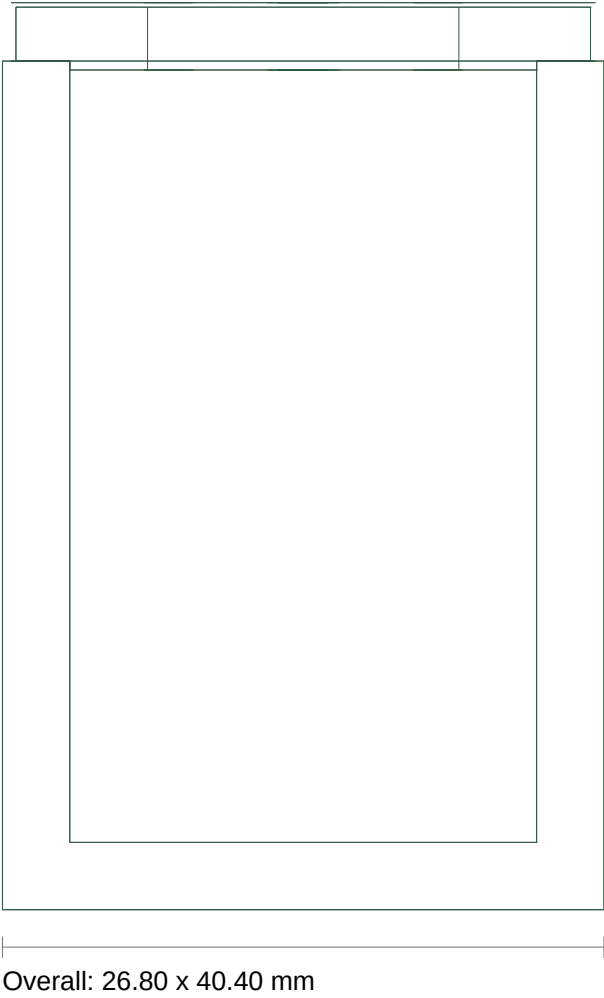
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 32.4 / 26.8 / 40.4 mm

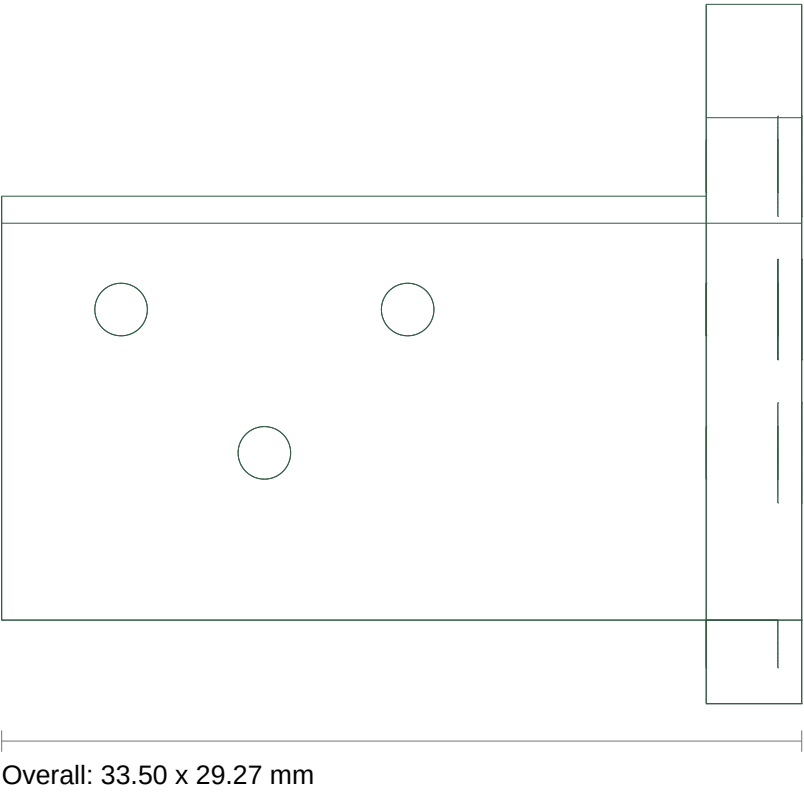
Material volume: 13731.2 mm3 | PLA solid-equivalent: 17.03 g (slicer mass will be lower)

Hip yaw horn plate on a U-channel around the hip roll servo. Print on the plate face; every wall is vertical.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

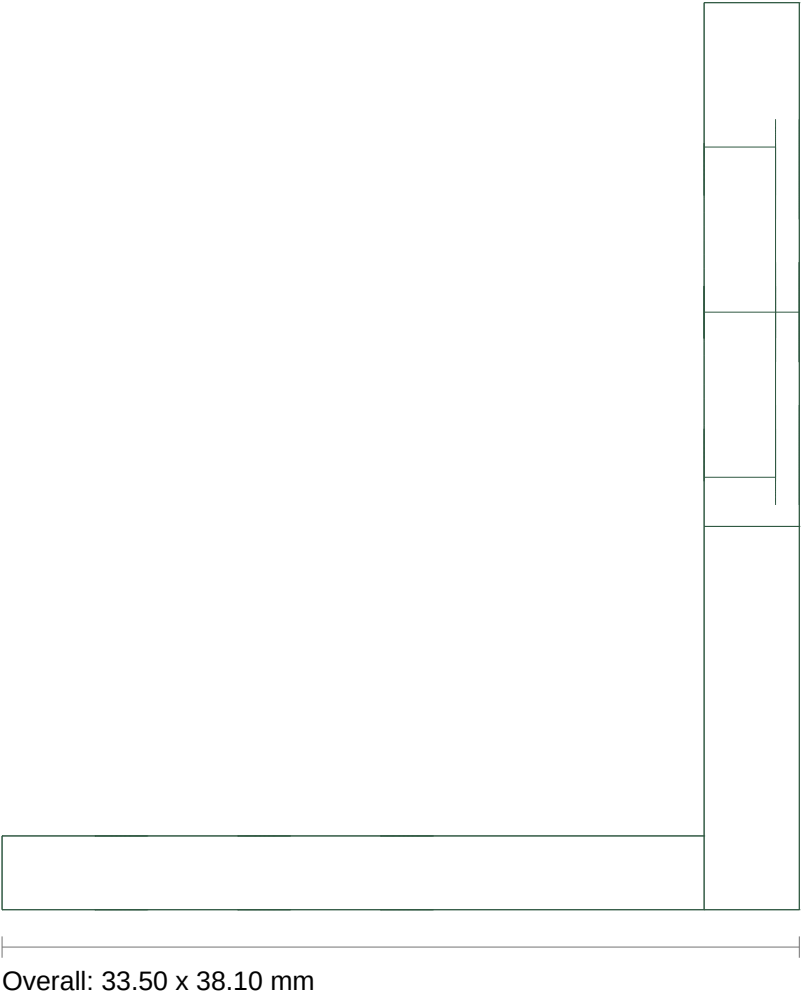
MICRO CAT Rev 1 / right_hip

link: right_hip | millimetres | original design, prototype reference, not production released

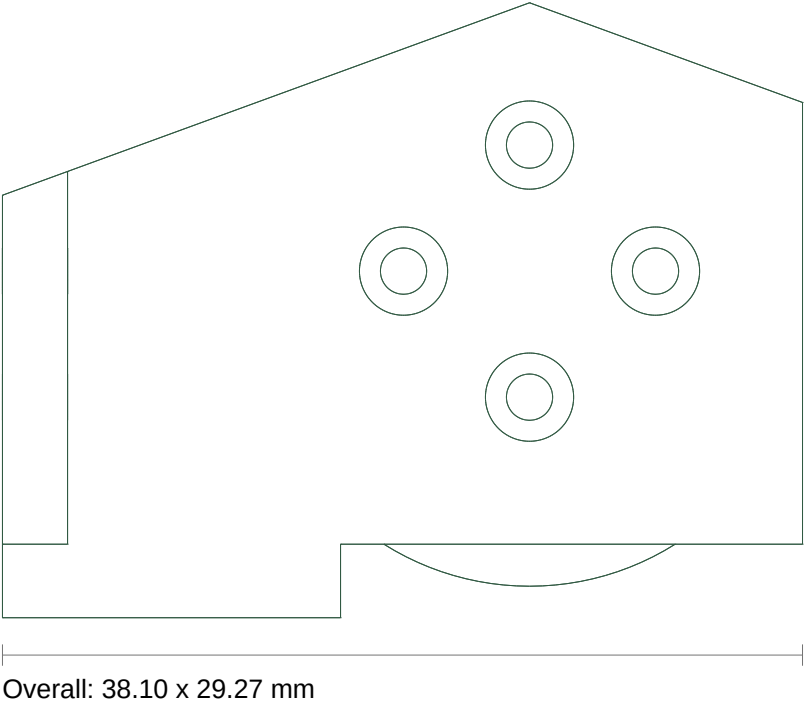
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 33.5 / 38.1 / 29.27 mm

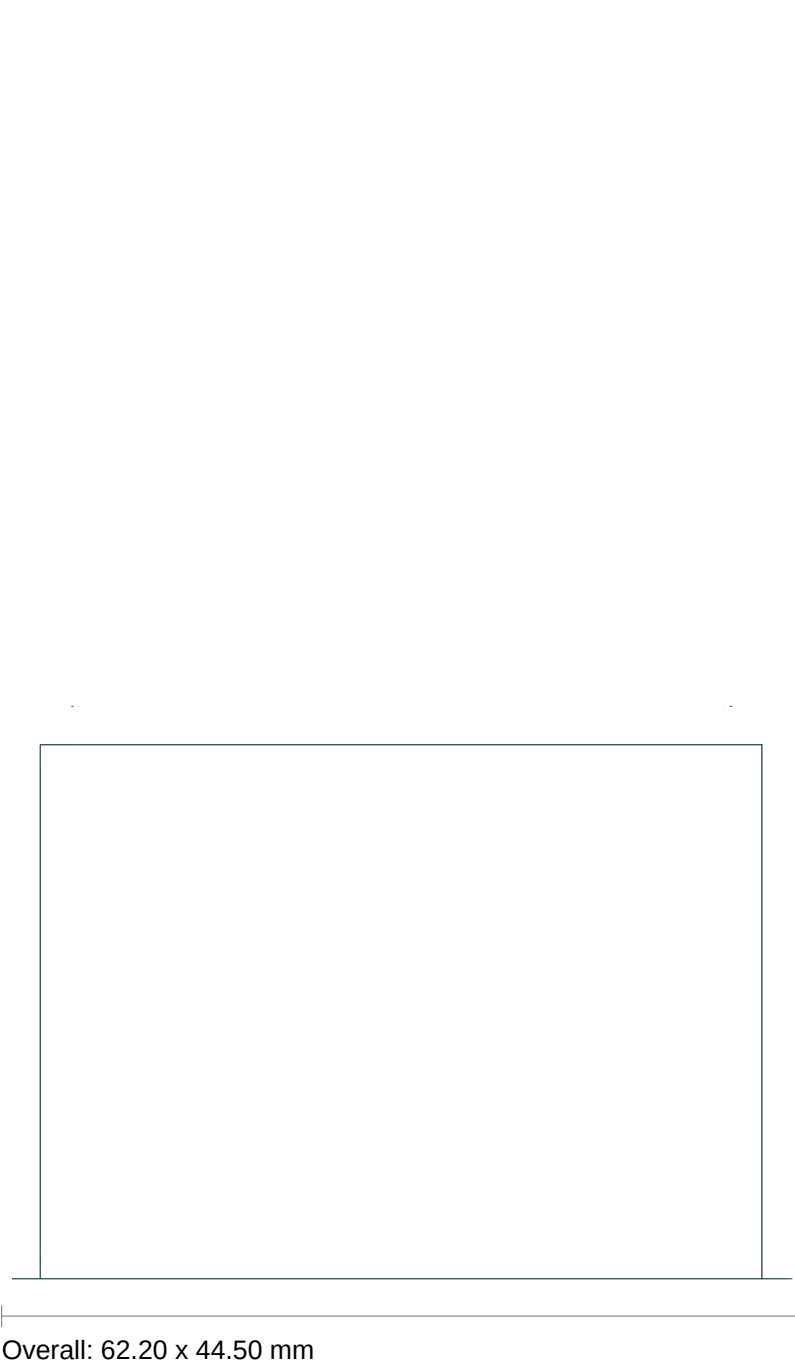
Material volume: 5061.1 mm3 | PLA solid-equivalent: 6.28 g (slicer mass will be lower)

Hip corner bracket: roll horn plate and pitch horn plate joined by one rib. Print on the pitch plate face.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

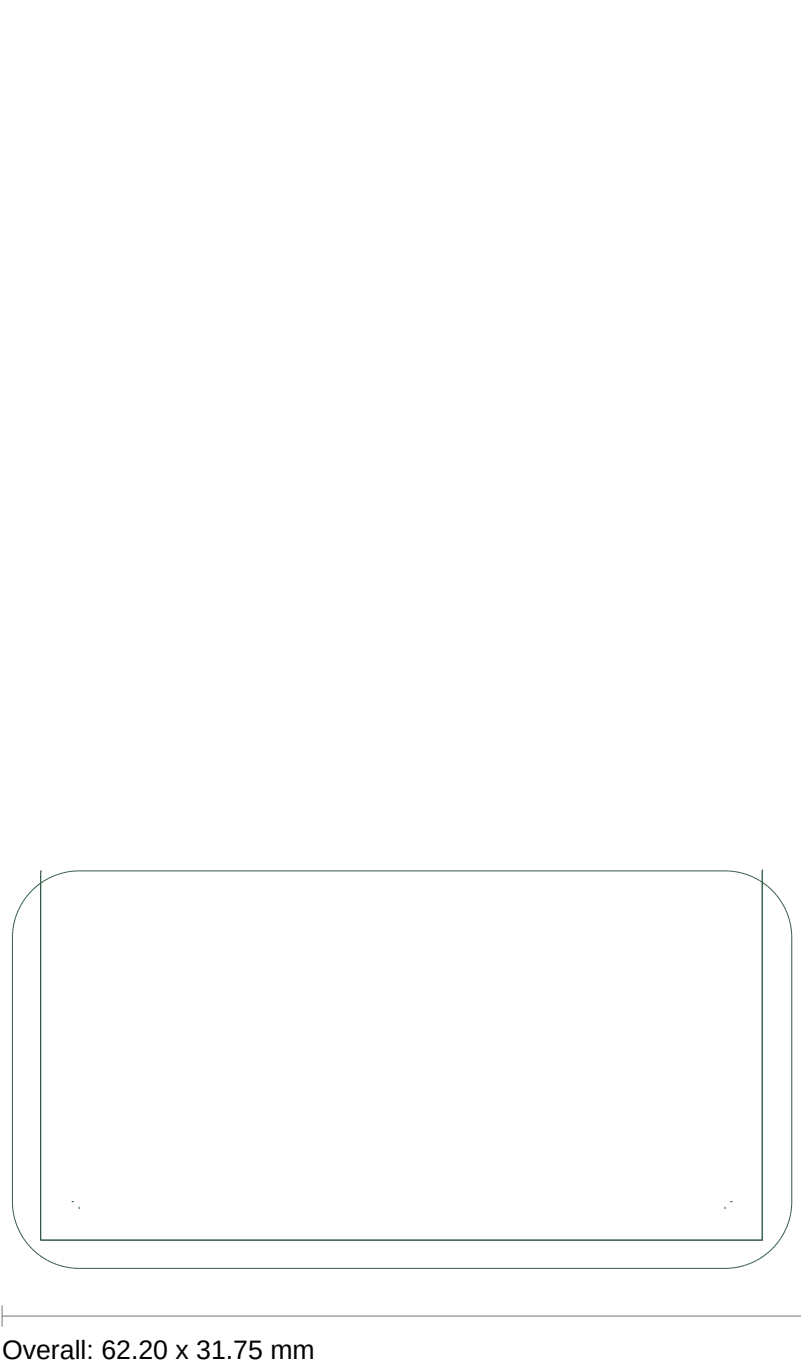
MICRO CAT Rev 1 / right_upper_leg

link: right_upper_leg | millimetres | original design, prototype reference, not production released

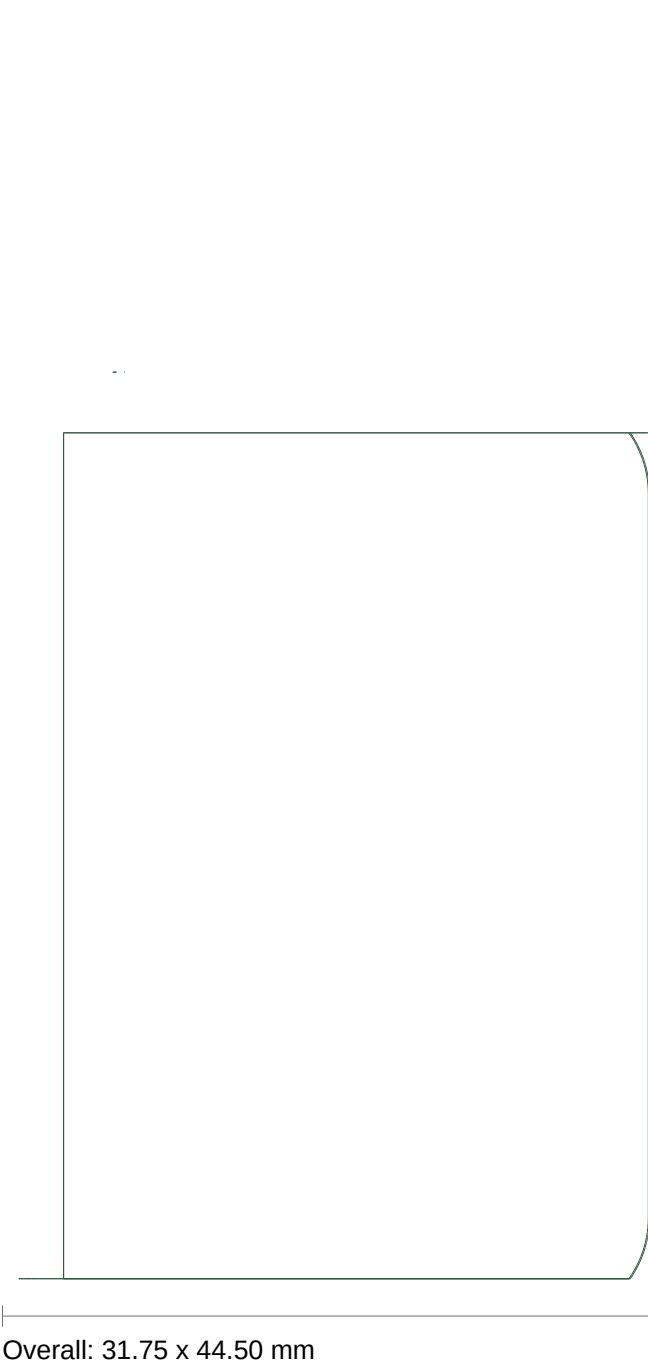
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 62.2 / 31.75 / 44.5 mm

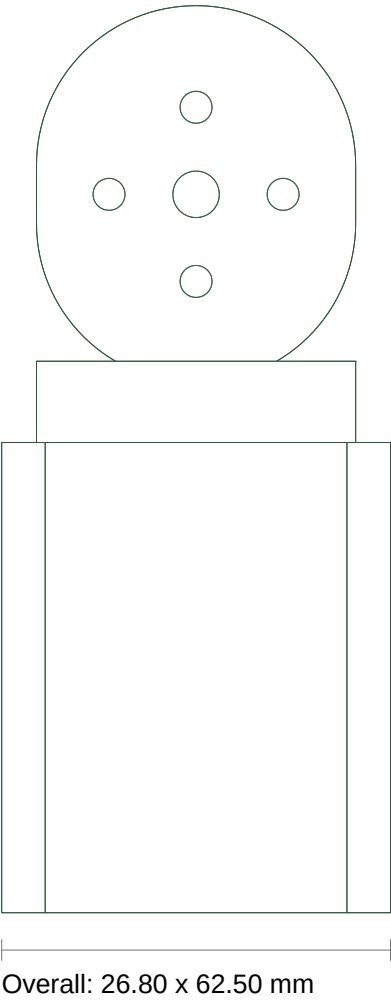
Material volume: 18593.5 mm3 | PLA solid-equivalent: 23.06 g (slicer mass will be lower)

Thigh shell around the hip pitch and knee servos, open on the inboard face. Print on the outboard face; the cavity opens upward, no support.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

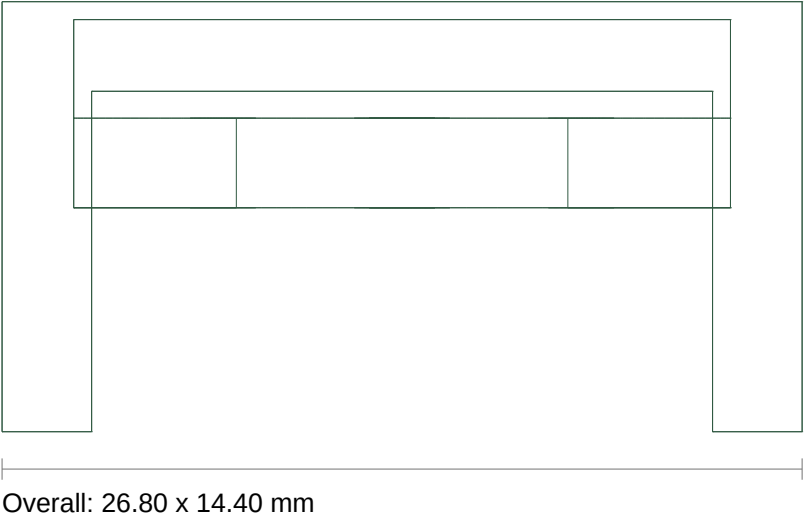
MICRO CAT Rev 1 / right_lower_leg

link: right_lower_leg | millimetres | original design, prototype reference, not production released

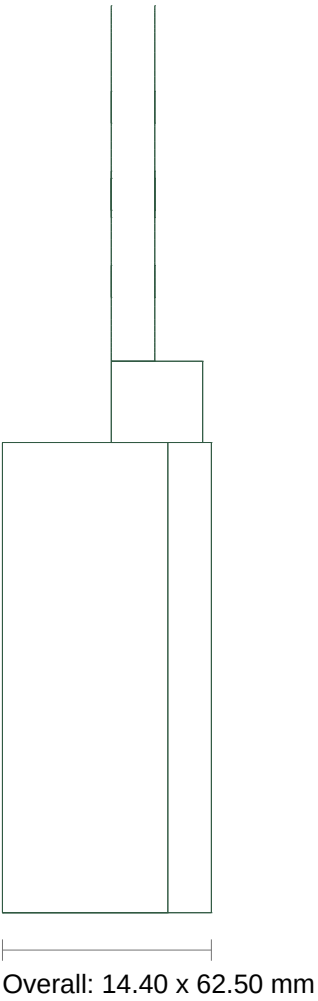
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 26.8 / 14.4 / 62.5 mm

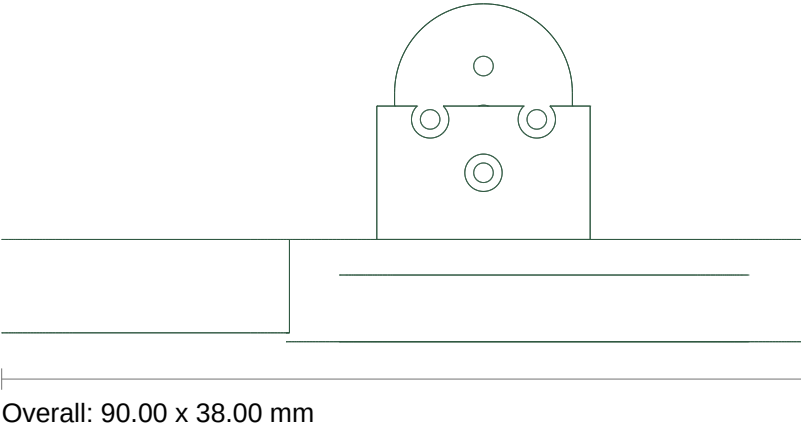
Material volume: 6898.7 mm3 | PLA solid-equivalent: 8.55 g (slicer mass will be lower)

Shin: knee horn plate stepping down into a channel around the ankle servo. Print on the inboard face.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

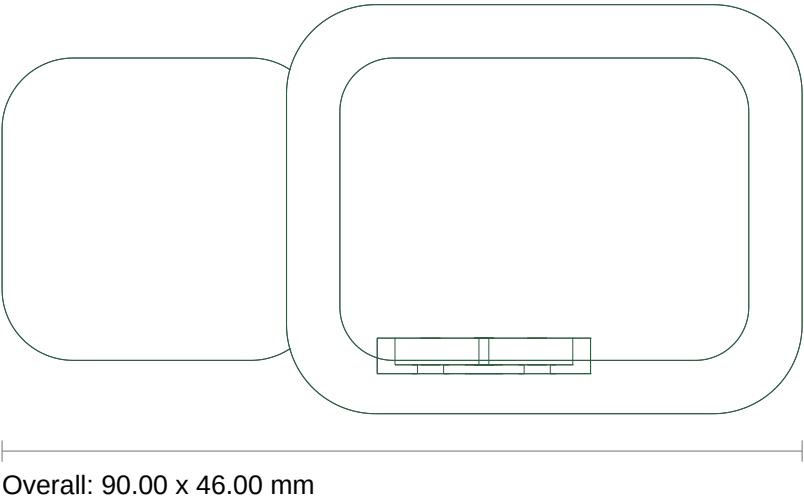
MICRO CAT Rev 1 / right_foot

link: right_foot | millimetres | original design, prototype reference, not production released

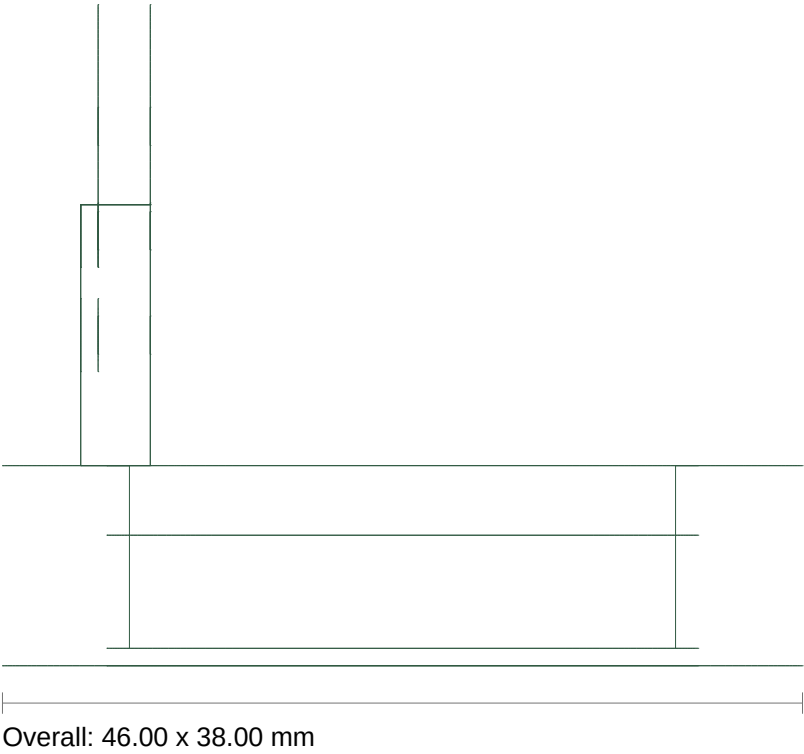
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 90.0 / 46.0 / 38.0 mm

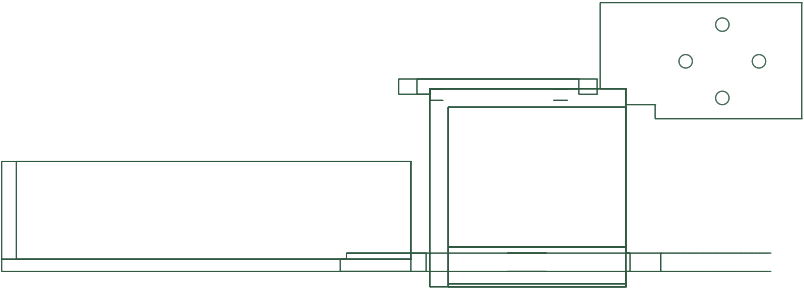
Material volume: 31227.4 mm3 | PLA solid-equivalent: 38.72 g (slicer mass will be lower)

Paw: ankle horn plate on a rounded sole, hollow underneath for a replaceable pad, plus a heel spur that sits 1.5 mm above the sole plane and reaches 24 mm behind it. Print sole down; the spur underside is a bridge 1.5 mm off the bed. Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO CAT Rev 1 / chassis

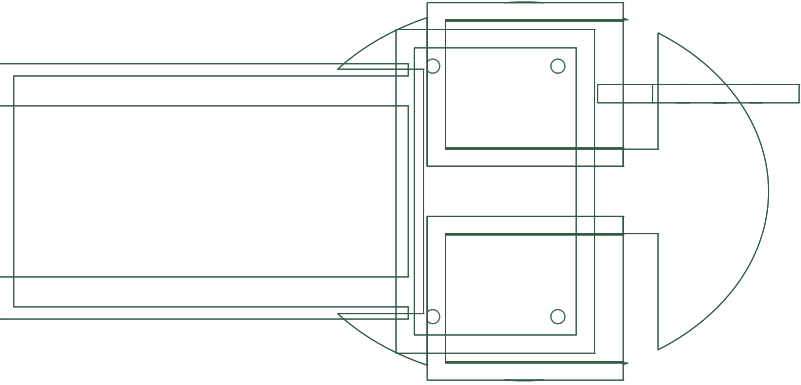
link: trunk | millimetres | original design, prototype reference, not production released

SIDE X-Z



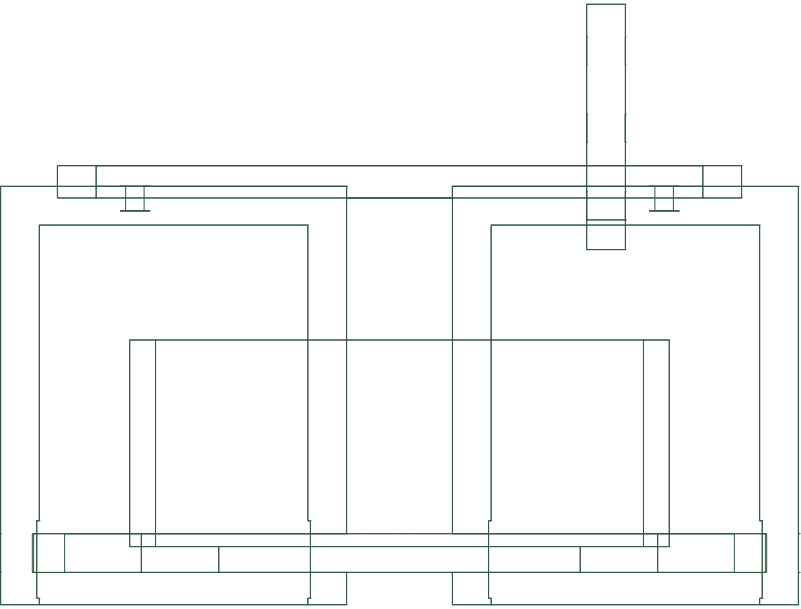
Overall: 131.00 x 46.50 mm

TOP X-Y



Overall: 131.00 x 61.98 mm

FRONT Y-Z



Overall: 61.98 x 46.50 mm

Assembly-frame extents X/Y/Z: 131.0 / 61.98 / 46.5 mm

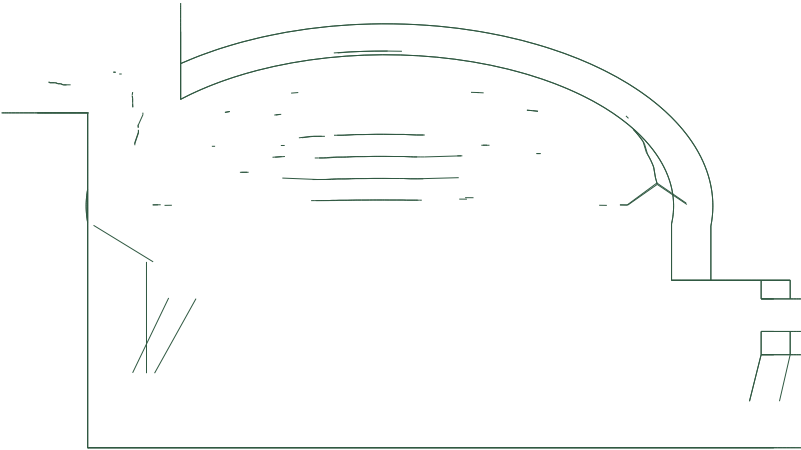
Material volume: 35653.3 mm3 | PLA solid-equivalent: 44.21 g (slicer mass will be lower)

Chassis: floor with two hip-yaw servo boxes, neck horn plate, battery tray and compute board wall. Print floor down; the servo boxes bridge 21 mm. Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO CAT Rev 1 / torso_shell

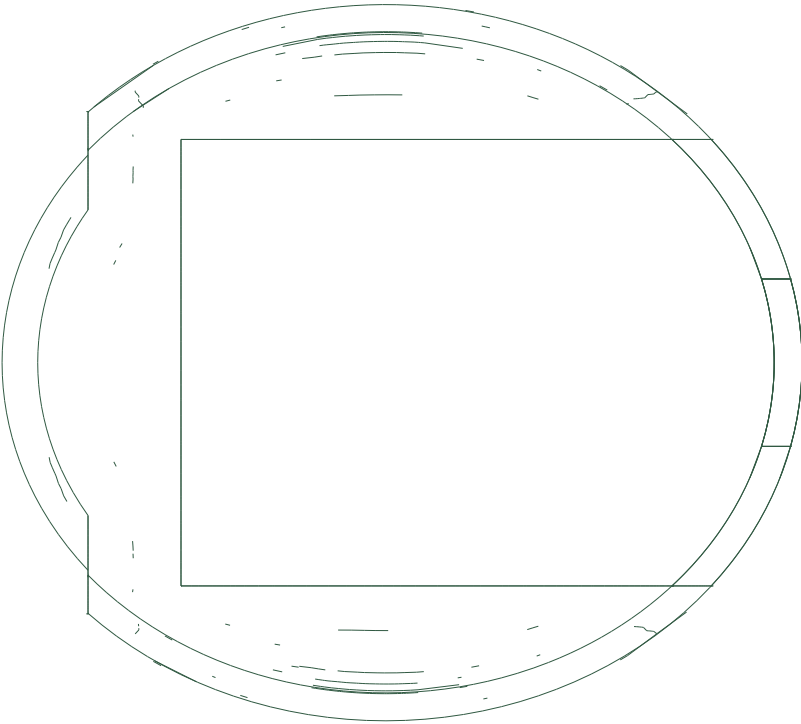
link: trunk | millimetres | original design, prototype reference, not production released

SIDE X-Z



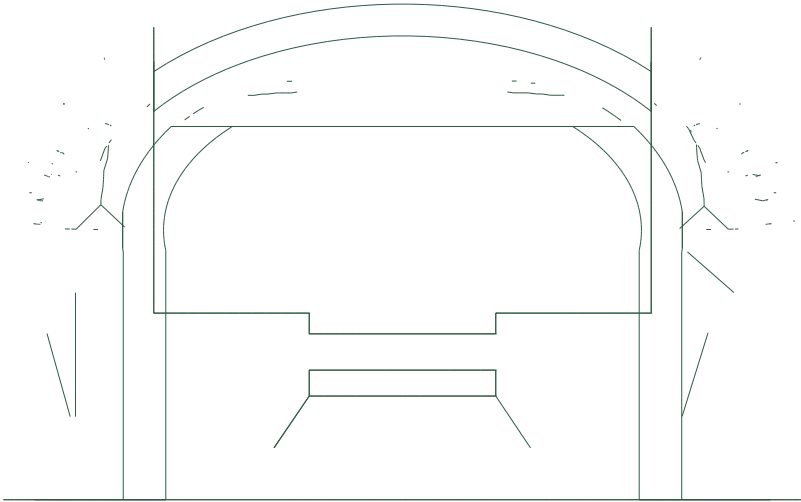
Overall: 86.04 x 47.80 mm

TOP X-Y



Overall: 86.04 x 77.20 mm

FRONT Y-Z



Overall: 77.20 x 47.80 mm

Assembly-frame extents X/Y/Z: 86.04 / 77.2 / 47.8 mm

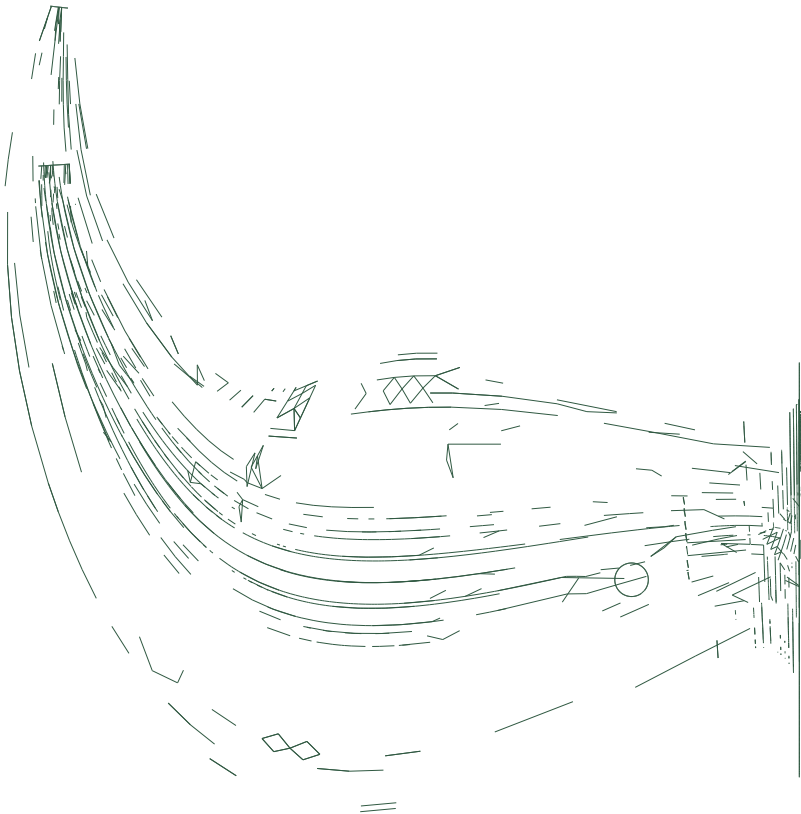
Material volume: 23445.0 mm3 | PLA solid-equivalent: 29.07 g (slicer mass will be lower)

One-piece egg shell with a chest grille, neck slot and tail opening, open under the hips. Four M3 through the flanks into chassis bosses. Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO CAT Rev 1 / tail_cover

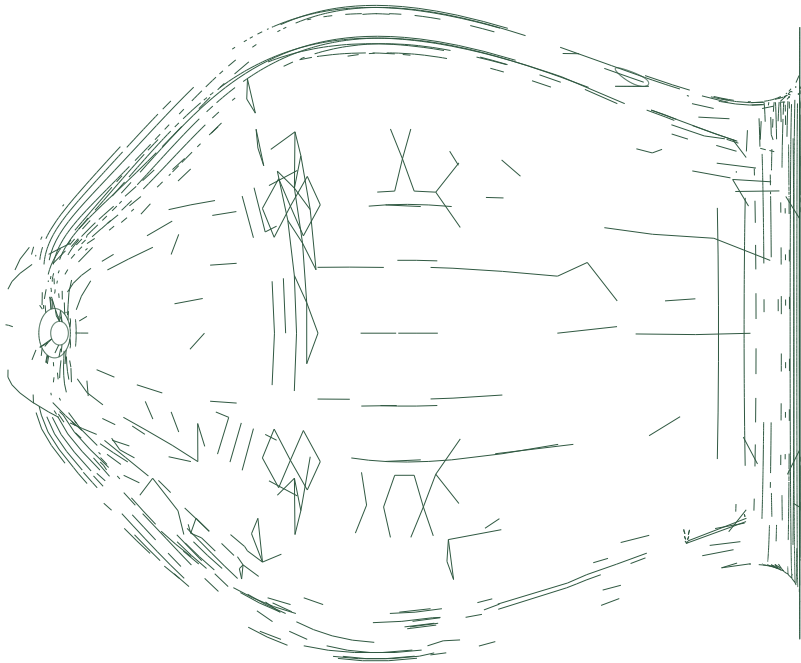
link: trunk | millimetres | original design, prototype reference, not production released

SIDE X-Z



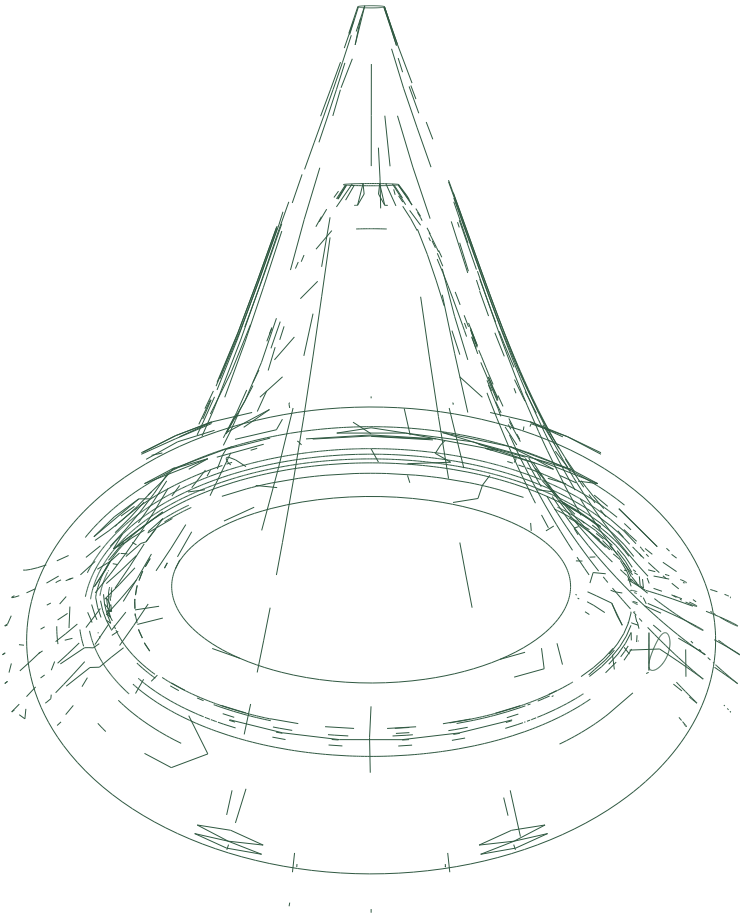
Overall: 81.06 x 81.59 mm

TOP X-Y



Overall: 81.06 x 66.38 mm

FRONT Y-Z



Overall: 66.38 x 81.59 mm

Assembly-frame extents X/Y/Z: 81.06 / 66.38 / 81.59 mm

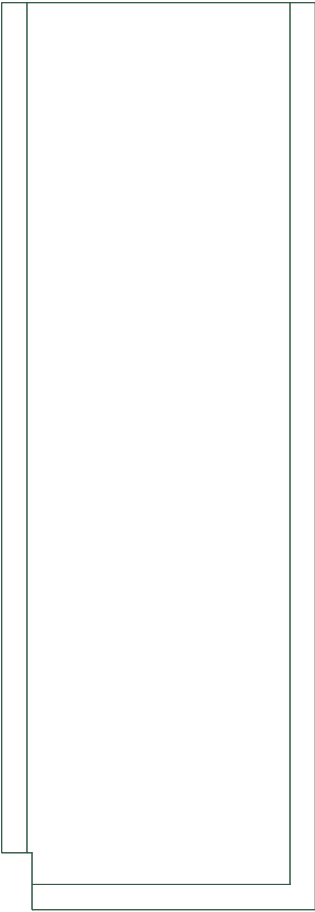
Material volume: 34323.4 mm3 | PLA solid-equivalent: 42.56 g (slicer mass will be lower)

Tail: curls up behind the body and doubles as the battery cover. Slides over the tray and takes two M3 into it.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO CAT Rev 1 / neck_link

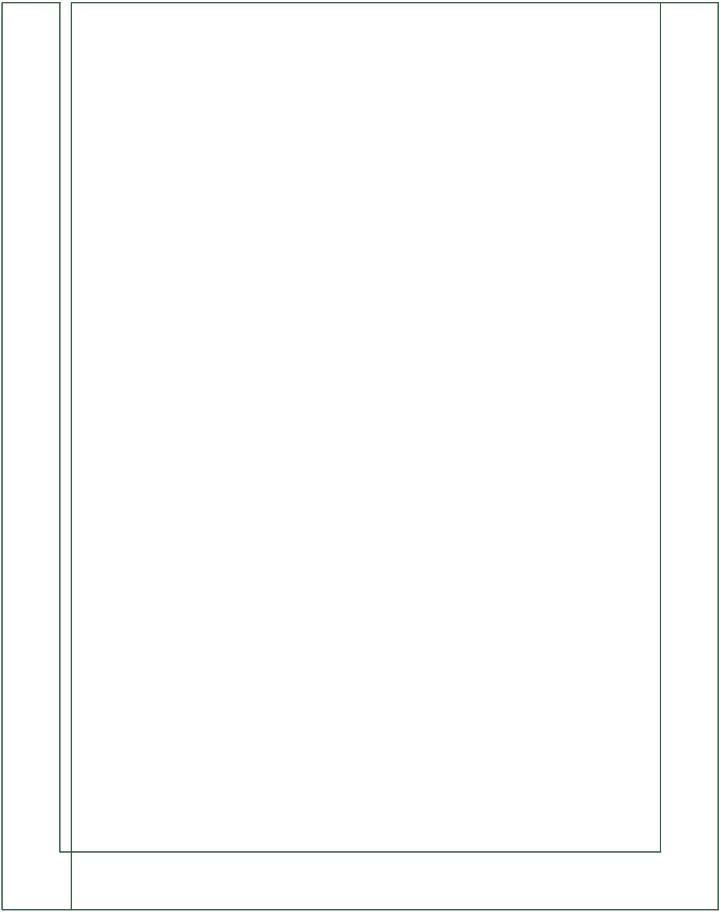
link: neck | millimetres | original design, prototype reference, not production released

SIDE X-Z



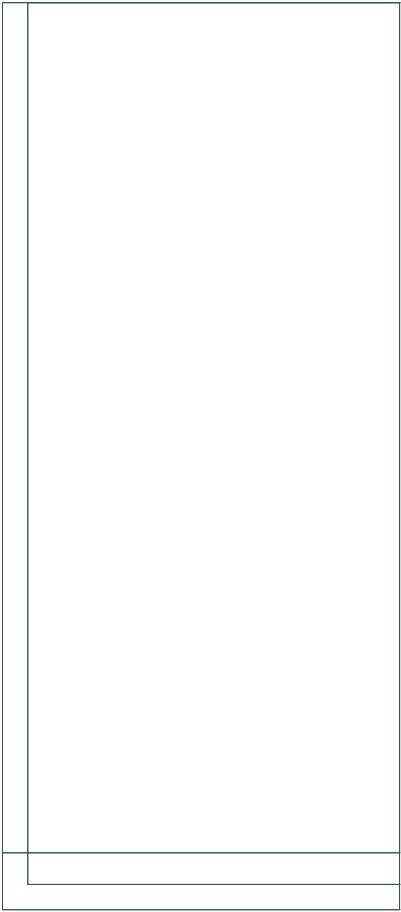
Overall: 24.80 x 71.70 mm

TOP X-Y



Overall: 24.80 x 31.40 mm

FRONT Y-Z



Overall: 31.40 x 71.70 mm

Assembly-frame extents X/Y/Z: 24.8 / 31.4 / 71.7 mm

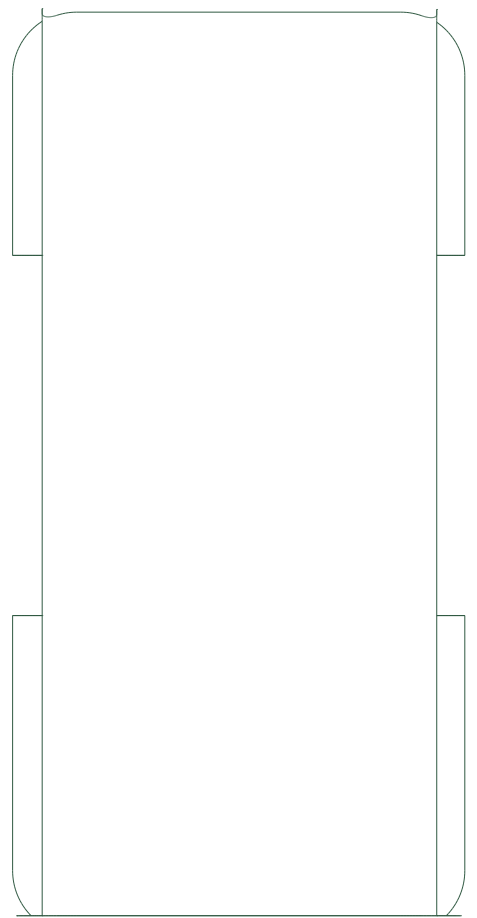
Material volume: 12901.6 mm3 | PLA solid-equivalent: 16.0 g (slicer mass will be lower)

Neck tube around the stacked neck-pitch and head-pitch servos, open on the horn side. Print on the closed face.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO CAT Rev 1 / neck_sleeve

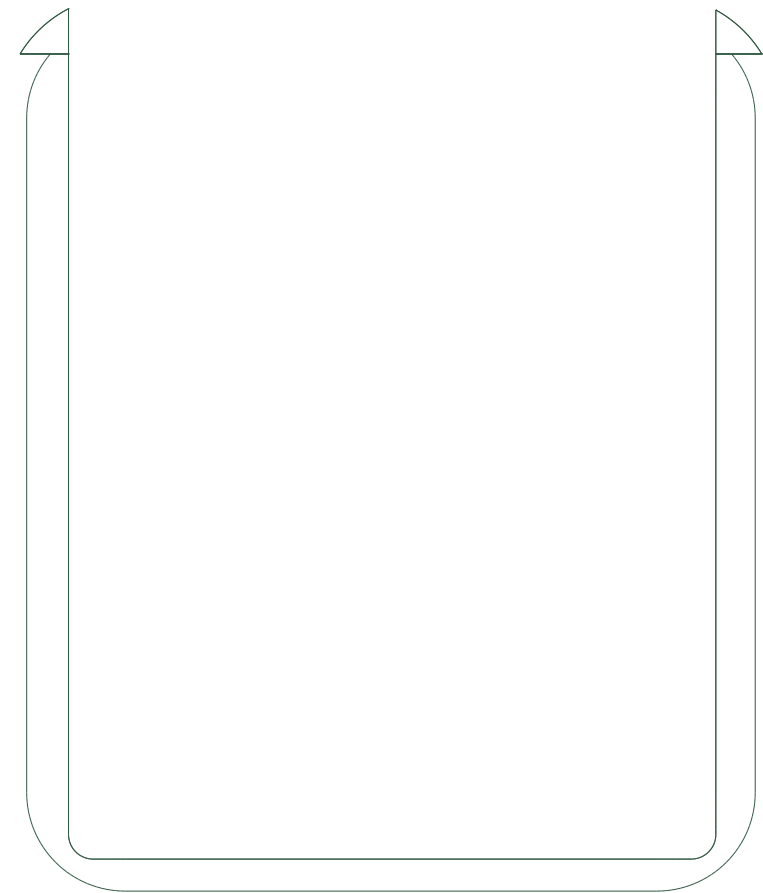
link: neck | millimetres | original design, prototype reference, not production released

SIDE X-Z



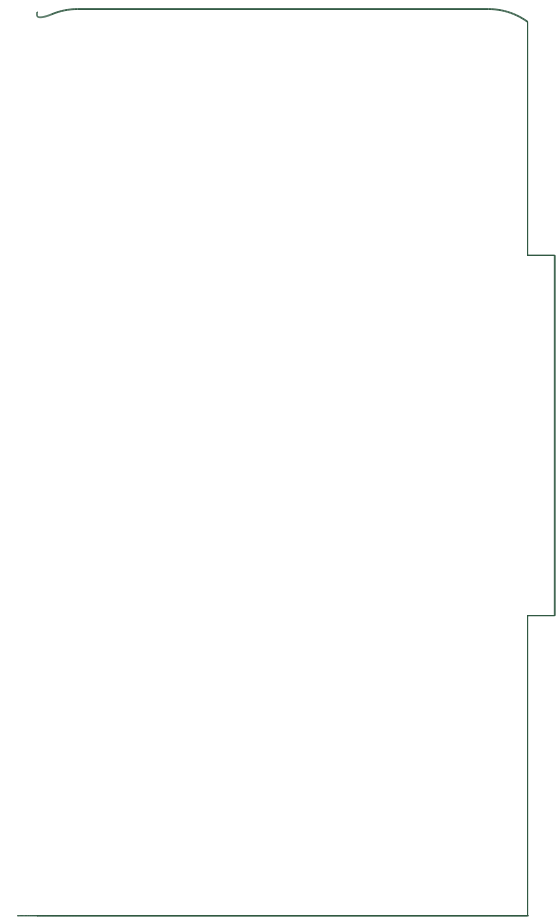
Overall: 31.60 x 60.44 mm

TOP X-Y



Overall: 31.60 x 36.84 mm

FRONT Y-Z



Overall: 36.84 x 60.44 mm

Assembly-frame extents X/Y/Z: 31.6 / 36.84 / 60.44 mm

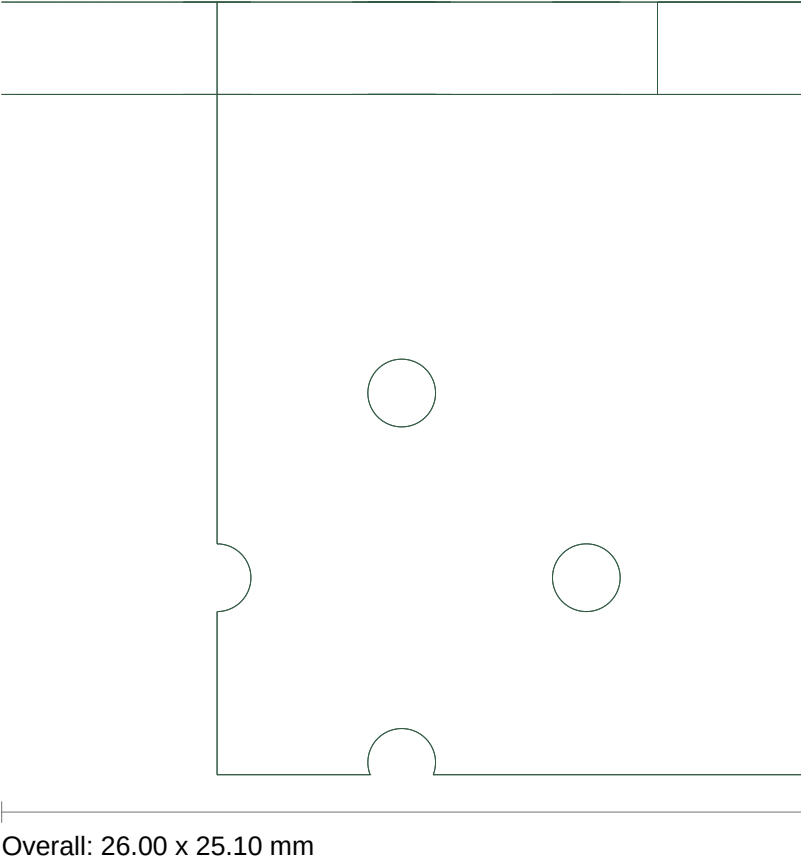
Material volume: 13852.3 mm3 | PLA solid-equivalent: 17.18 g (slicer mass will be lower)

Neck cover: rounded shell over the neck column, open on the horn side; pitches with the neck.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

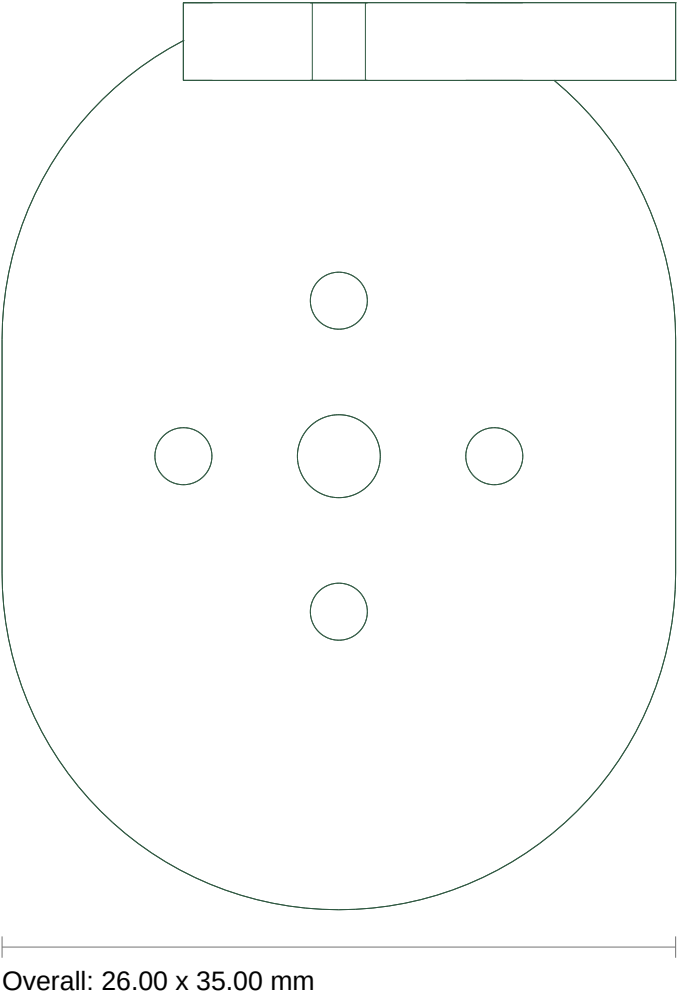
MICRO CAT Rev 1 / head_base

link: head_base | millimetres | original design, prototype reference, not production released

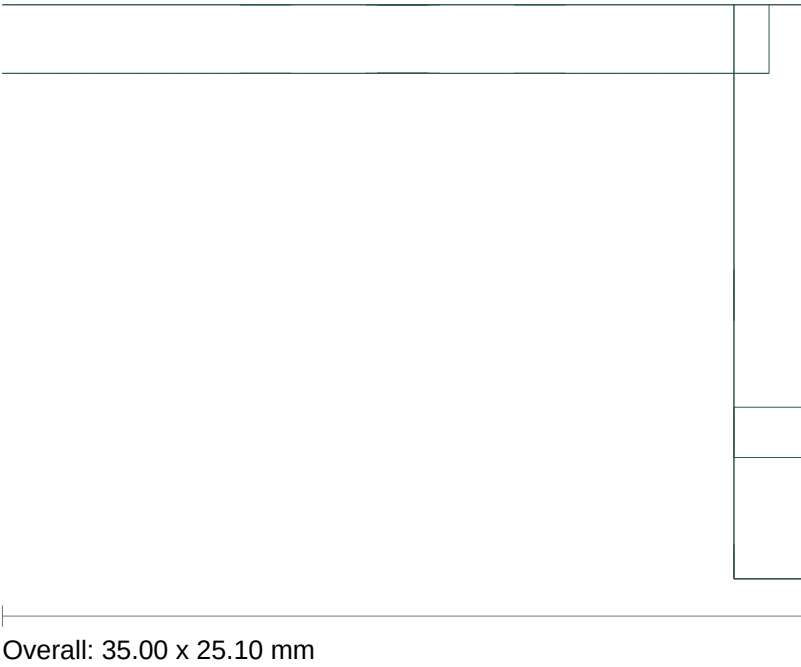
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 26.0 / 35.0 / 25.1 mm

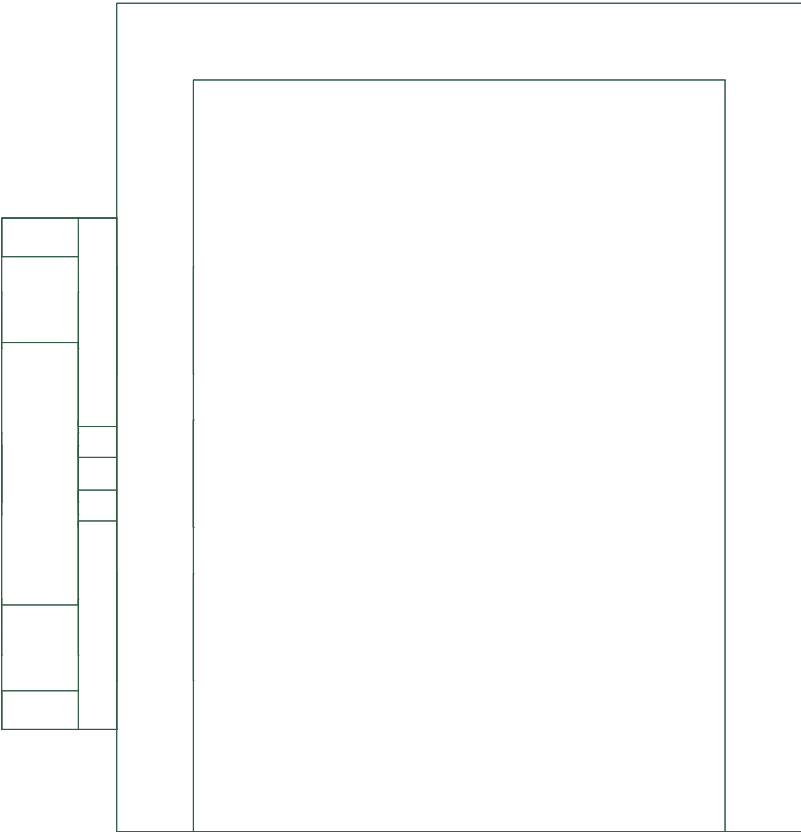
Material volume: 3523.0 mm3 | PLA solid-equivalent: 4.37 g (slicer mass will be lower)

Head base: head-pitch horn plate joined to the head-yaw horn plate. The only part that stays still while the head yaws.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO CAT Rev 1 / head_yoke

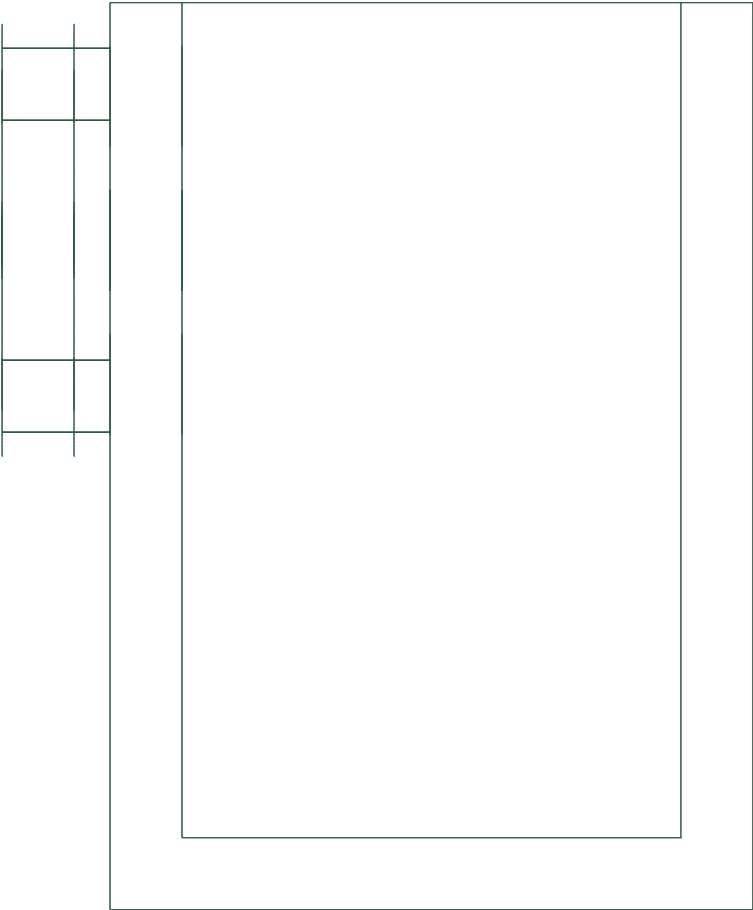
link: head_yoke | millimetres | original design, prototype reference, not production released

SIDE X-Z



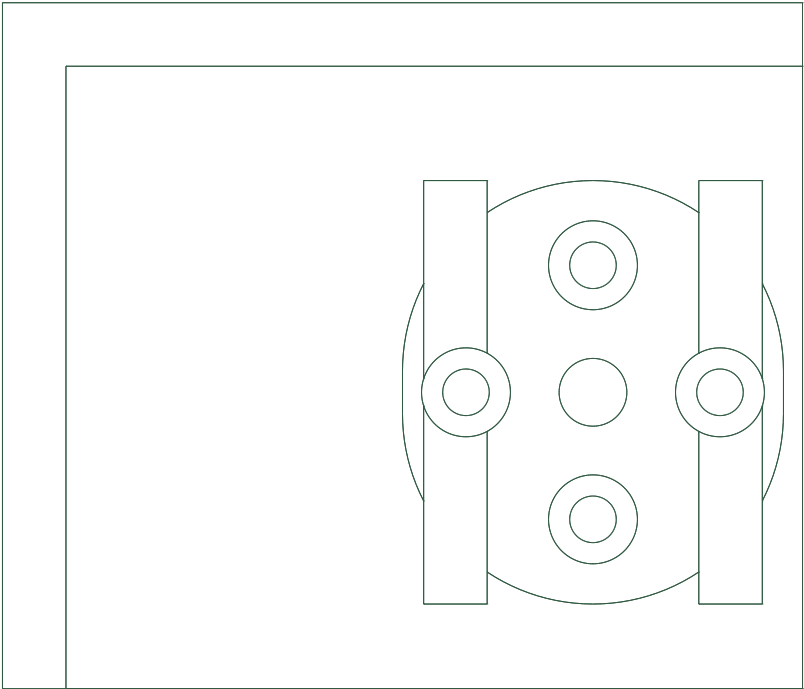
Overall: 31.30 x 32.40 mm

TOP X-Y



Overall: 31.30 x 37.80 mm

FRONT Y-Z



Overall: 37.80 x 32.40 mm

Assembly-frame extents X/Y/Z: 31.3 / 37.8 / 32.4 mm

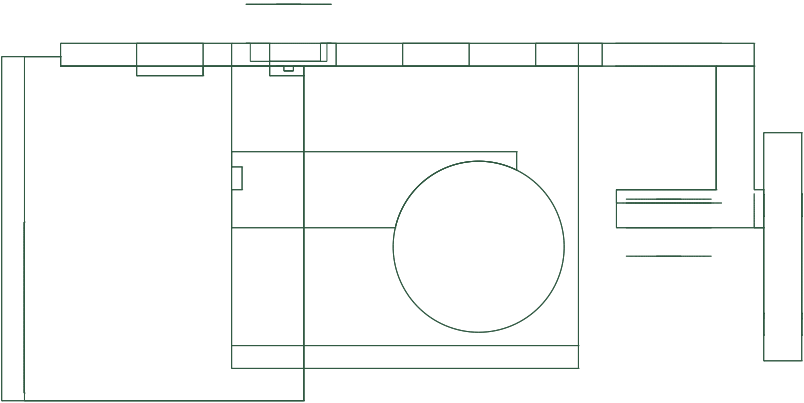
Material volume: 12429.4 mm3 | PLA solid-equivalent: 15.41 g (slicer mass will be lower)

Yoke: box around the head-yaw servo with two ribs out to the head-roll horn plate.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO CAT Rev 1 / head_frame

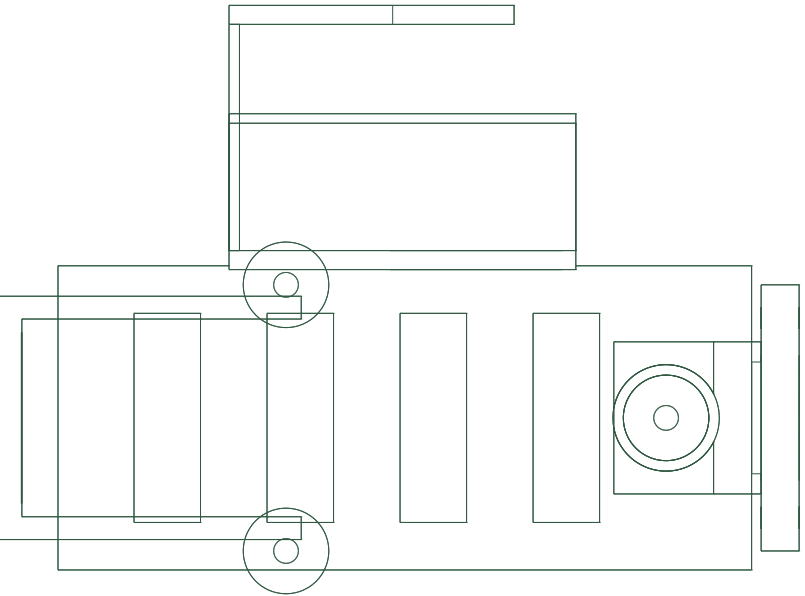
link: head | millimetres | original design, prototype reference, not production released

SIDE X-Z



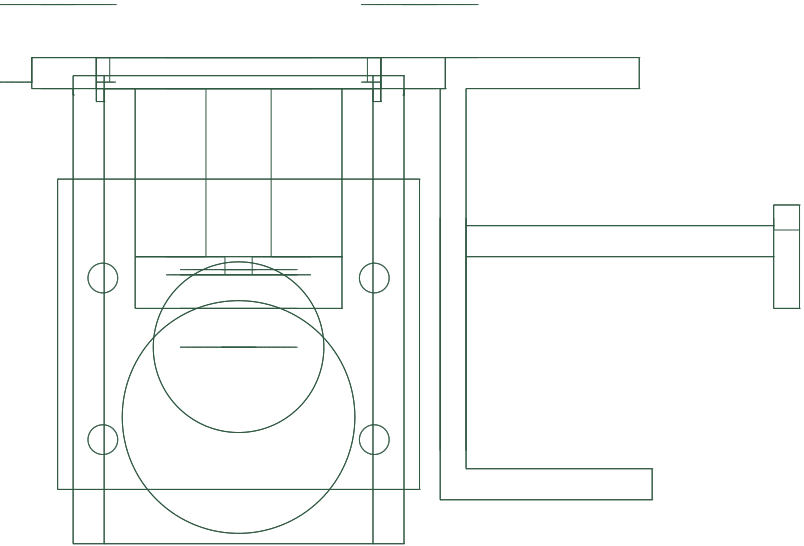
Overall: 84.20 x 41.70 mm

TOP X-Y



Overall: 84.20 x 61.90 mm

FRONT Y-Z



Overall: 61.90 x 41.70 mm

Assembly-frame extents X/Y/Z: 84.2 / 61.9 / 41.7 mm

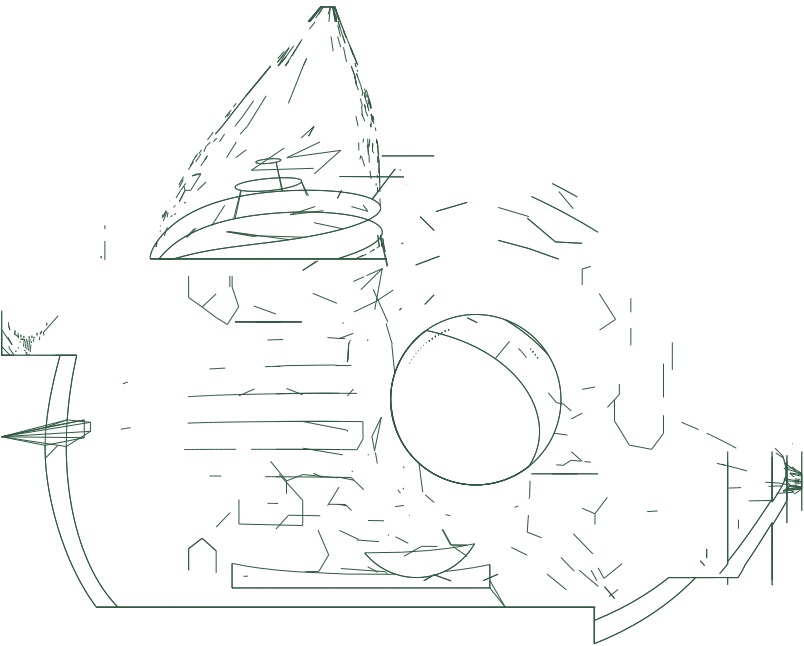
Material volume: 20253.1 mm3 | PLA solid-equivalent: 25.11 g (slicer mass will be lower)

Head frame: one shell with a deck over the yoke, the head-roll and jaw servo bays, a forward beam, the camera wall and three skull bosses.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO CAT Rev 1 / skull

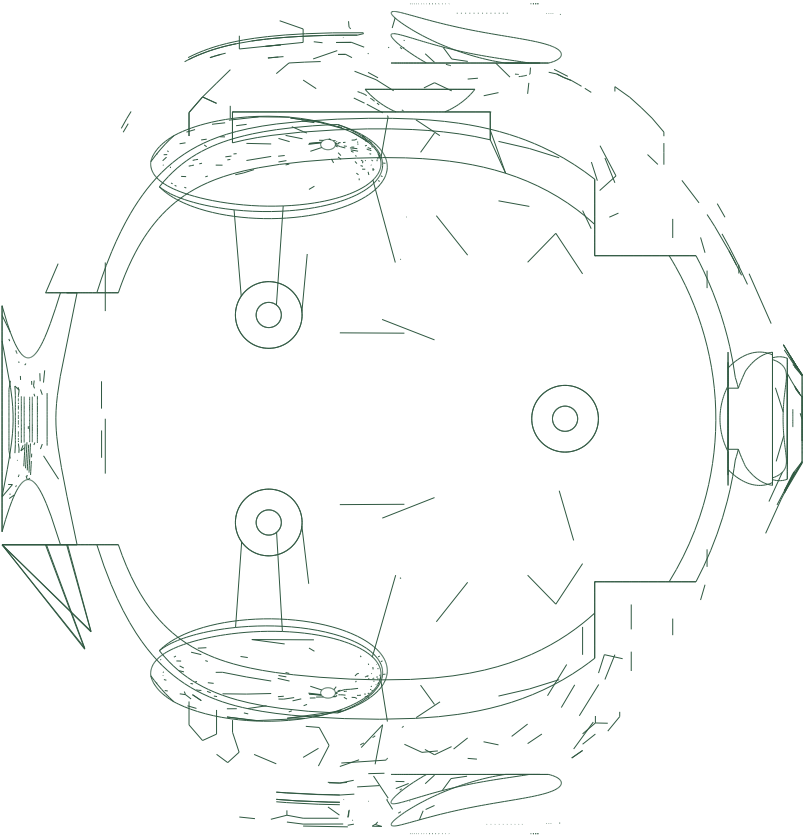
link: head | millimetres | original design, prototype reference, not production released

SIDE X-Z



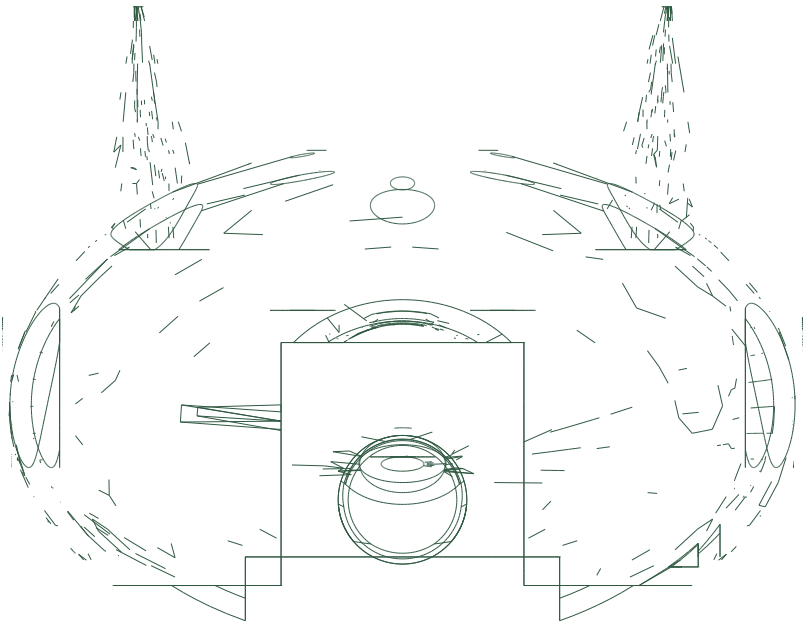
Overall: 108.00 x 85.93 mm

TOP X-Y



Overall: 108.00 x 112.00 mm

FRONT Y-Z



Overall: 112.00 x 85.93 mm

Assembly-frame extents X/Y/Z: 108.0 / 112.0 / 85.93 mm

Material volume: 69762.2 mm3 | PLA solid-equivalent: 86.51 g (slicer mass will be lower)

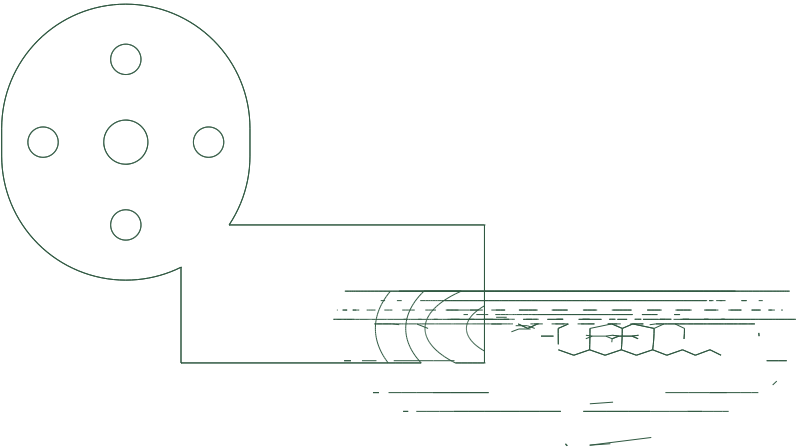
Head: one-piece round cat cranium with a short muzzle, solid ears and a nose lens hood. Three M3 down into the head frame; eyes, ear insides and nose are painted, not parts.

Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO CAT Rev 1 / jaw_chin

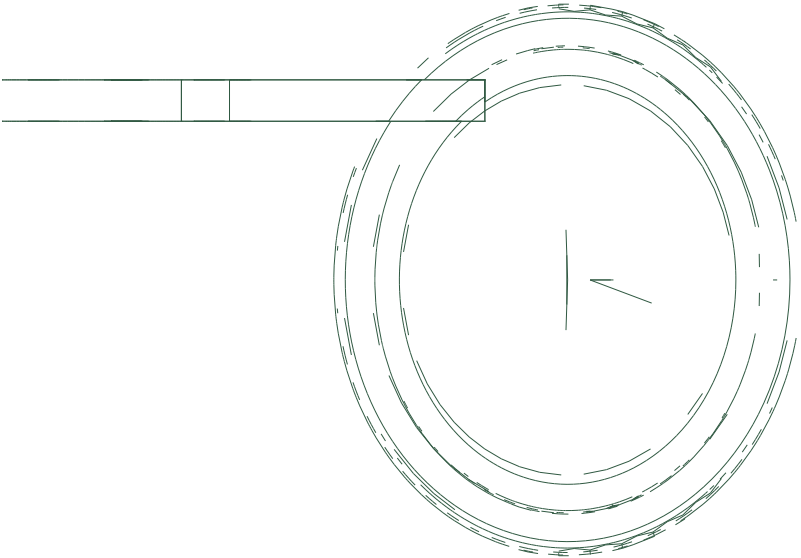
link: jaw | millimetres | original design, prototype reference, not production released

SIDE X-Z



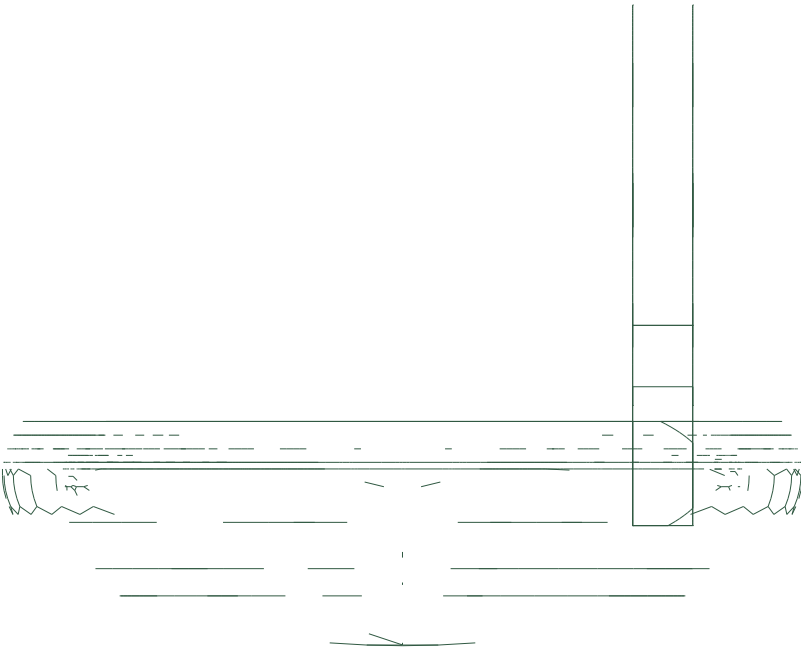
Overall: 58.00 x 32.00 mm

TOP X-Y



Overall: 58.00 x 39.97 mm

FRONT Y-Z



Overall: 39.97 x 32.00 mm

Assembly-frame extents X/Y/Z: 58.0 / 39.97 / 32.0 mm

Material volume: 5602.3 mm3 | PLA solid-equivalent: 6.95 g (slicer mass will be lower)

Lower jaw: rounded chin cup on a single arm bolted to the jaw servo horn.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.